

INDIAN INSTITUTE OF TECHNOLOGY, BOMBAY

DUAL DEGREE THESIS

Microscallop Modelling, Motion Planning and Control

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for the degree of Integrated B.Tech and M.Tech*

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Declaration of Authorship

I, Yashaswini MURTHY, declare that this thesis titled, “Microscallop Modelling, Motion Planning and Control” and the work presented in it are my own. I confirm that:

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Abstract

Department of Mechanical Engineering

Integrated B.Tech and M.Tech

Microscallop Modelling, Motion Planning and Control

by Yashaswini MURTHY

This thesis makes two contributions. The first part studies model-based predictions of a microscallop moving in a non-Newtonian fluid in the low Reynolds regime[LRN]. The larger objective here is that a study of such motion would provide insight into understanding manipulation and control in biological applications at such micro and nano scales. A Lagrangian approach is adopted with certain aspects of fluid dynamics, like the viscous forces, being incorporated from existing empirical relationships along with the underlying assumptions. This model is then simulated for different initial conditions and key parameters of the non-Newtonian fluid, and the results are corroborated with a few existing experimental results on a similar mechanism under identical conditions. The dynamics of the mechanism being very complex, a few model simplifications are introduced in order to explore control-relevant features such as *accessibility* of the system at certain configurations.

The second part concerns the coordinated control of two microscallops in Newtonian fluids. The results in this part rest on the *no-many scallop theorem*, which states that multiple scallops with non-trivial phase differences can have non zero displacements when subject to periodic actuation. A kinematic model of the two scallop system, accounting for the hydrodynamic interactions used to overcome the reciprocal theorem is developed. Numerical experiments relevant to certain test cases are performed in order to validate the model. Then the controllability of the system is studied, following which Lyapunov functions are constructed to achieve the desired configurations in the coordinated motion. A differential geometric method for motion planning is then adopted to compute feasible paths to steer the system from one configuration to another in a specified time.

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List of Abbreviations

LRN	Low Reynolds Number
NN	Non Newtonian
LARC	Lie Algebra Rank Condition
STLC	Small Time Locally Controllable

List of Symbols

μ	coefficient of viscosity	Pa s
b_c	drag coefficient (non Newtonian case)	Pa s
k	drag coefficient (Newtonian)	Pa s
l	length of scallop link	m
a	thickness of scallop link	m
λ	relaxation time	s
n	shear thinning coefficient (unitless)	
R_c	correction factor of drag coefficients (unitless)	

Dedicated to my parents

Chapter 1

Introduction

With the primary intention of making the life of humans easier, and performing tasks which are hazardous to people, the field of robotics has witnessed a drastic technological revolution ever since its beginning. It is now routine to see robot manipulators being used for welding and painting car bodies on assembly lines, stuffing printed circuit boards with IC components, inspecting and repairing structures in nuclear, undersea, and underground environments, and even picking oranges and harvesting grapes in agriculture.



FIGURE 1.1: The Unimation PUMA (Programmable Universal Manipulator for Assembly) used for CNC in 1961(Murray et al., 1994)

In manufacturing, Computer Numerically Controlled (CNC) machine tools for accurate milling of low-volume, high-performance aircraft parts were also developed(Murray et al., 1994). Following this, robots with sensory feedback were developed. The potential range of robotic applications requires different types of sensors to perform different kinds of sensing tasks. Specialized devices have been developed to meet various sensing needs such as orientation, displacement, velocity, acceleration and force(Hall, 1984).

Walking robotics research also gathered a new momentum due to the new inventions in mechanisms, material science, electronics, control system, and computers. This laid the foundation of modern walking robotics research for the systematic design and development of walking machines for their intended applications in rough terrain(Kenzo Nonami, 2013).

However, in the last decade of the 20th century, Microbots were spurred by the microcontroller and the appearance of miniature mechanical systems on silicon (MEMS), although many microbots do not use silicon for mechanical components other than sensors. As a further advancement, due to their high utility in biomedical applications, there has been an increased interest in micro and nanorobots.

The ability to access small spaces inside the human body at LRN has facilitated

research into controllable micro and nanorobotics. Of particular interest to this work is drug delivery, which in itself is a topic with vast room for technological improvements along with many considerations and challenges needed to be addressed (Cheang and Kim, 2015). To aid in drug delivery and many other minimally invasive applications of microrobots it is important to understand the environment in which they operate. Since most of the fluids in the human body are of the NN kind, it is necessary to study the properties of such fluids at the LRN in order to model microrobots for the same.

1.1 Applications of microrobotics

Due to the minimally invasive nature of medical microrobots and the ability to precisely navigate these drug-carrying devices in bodily fluids, there exists great potential to introduce this new technology into medical practice. A few of the applications of the microrobots in the LRN regime are now enlisted.

- In recent years, there has been significant progress in robot assisted colonoscopy and in miniature robots for use in the gastrointestinal tract. Reducing the dimensions of microrobots would enable them to access additional locations in the human body such as the circulatory system, the urinary system, and the central nervous system through wireless means. Microrobots could navigate natural pathways, enabling intervention with minimal trauma (Nelson, Kaliakatsos, and Abbott, 2010).
- Magnetic microrobots that can be wirelessly steered using external magnetic fields present an exciting new dimension for targeted drug delivery in the eye. Microrobots would eliminate the need for a vitrectomy, and, in case of an adverse reaction, enable the removal of the microrobot, which is currently not possible with immobile intraocular implants (Fusco et al., 2014).

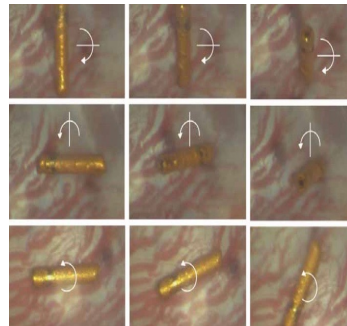


FIGURE 1.2: Rotation of the intravitreal microrobot in lapine vitreous (in vivo) around different axis (Fusco et al., 2014)

- Microrobots can be used for the localized delivery of chemical and biological substances, as well as various forms of energy. Brachytherapy is the placement of a radioactive source, sometimes referred to as a radioactive seed, near unwanted cells such as a tumor (Kassim et al., 2006). The radiated energy results in death of the cells close to the radioactive source (Nelson, Kaliakatsos, and Abbott, 2010).

- Biopsy or excision can also be performed by microrobots. In the simplest case, the microrobot retrieves a tissue sample and is removed from the body, and the analysis is conducted *ex vivo*. If biopsy or excision is combined with remote-sensing technology, the sample can be analyzed *in situ* (Nelson, Kaliakatsos, and Abbott, 2010).
- Ablation is the removal of material from the surface of an object. In its simplest form, this is accomplished by scraping. Microrobots may be particularly well suited for utilizing rotary motion to perform ablation tasks, which would be useful, for example, in the removal of fatty deposits from the internal walls of blood vessels. Another possibility is ultrasound ablation, in which a microrobot uses a resonating mechanical structure to emit ultrasonic pressure waves to destroy an object such as a kidney stone (Nelson, Kaliakatsos, and Abbott, 2010).
- Scaffolds act as cell support frames on which nerves can be regenerated, artificial organs developed, and blood vessels regrown. The microrobot itself can act as the scaffold, or the microrobot can deploy microscopic building blocks that act as scaffolding (Han et al., 2005), (Nelson, Kaliakatsos, and Abbott, 2010).
- Stents are structures that are used to keep passageways open. A typical example is the use of a stent to keep blood flowing through a clogged vessel. The microrobot itself would serve as the stent and would navigate and deploy in the location of interest (Nelson, Kaliakatsos, and Abbott, 2010).
- Occlusions can be introduced to intentionally block a passageway, either temporarily or permanently. Microrobots can function as occlusions, for example, to clog a blood vessel that nourishes a tumor (Nelson, Kaliakatsos, and Abbott, 2010).
- Implants or electrodes can be introduced by microrobots. A microrobot can act as a temporary or permanent implant, such as an electrode for brain stimulation. The electrode may operate wirelessly, or a wirelessly guided microrobot may tow a fine wire (Nelson, Kaliakatsos, and Abbott, 2010).
- Their application is significant in telemetry as well. Microrobots can perform remote sensing. For example, microrobots can transmit the time history of a physical signal of interest (e.g., oxygen concentration) or transmit a simple binary signal upon detecting the presence of an analyte of interest (e.g., cancer, blood) (Nelson, Kaliakatsos, and Abbott, 2010).

Most of the above applications require cooperative manipulation of these microrobots. Nature presents a myriad of such cooperative mechanisms exploited by microorganisms to ensure their basic survival. Depending on the external conditions, microorganisms exhibit various spacio-temporal patterns in coordination with one other, through chemotactic signalling and cell to cell interactions. The above applications can be summarised in the following figure.

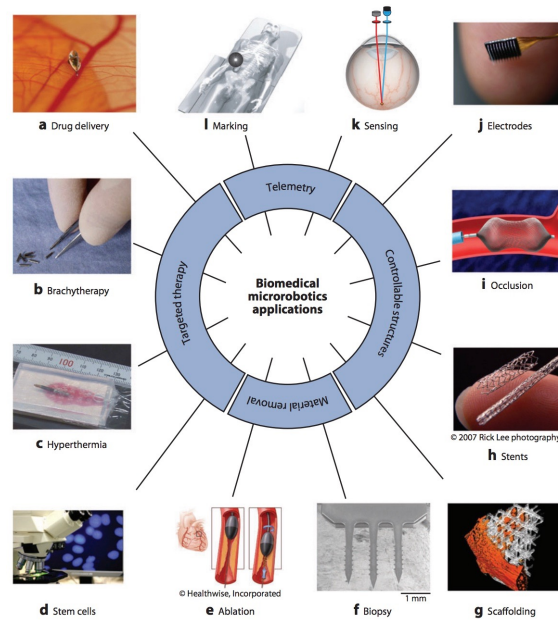


FIGURE 1.3: Various applications of microrobots in the biomedical industry(Nelson, Kaliakatsos, and Abbott, 2010)

By the virtue of their size and speeds, these organisms operate in an environment that is termed the LRN regime, which imposes certain constraints over locomotion; this shall be delineated later. The focus of this thesis is to illustrate and examine in detail some of the mechanisms relating to locomotion and present ways to manipulate locomotion to achieve desired goals (Ceylan et al., 2017).

Chapter 2

Fluid dynamics

2.1 Mathematical modelling

The study of locomotion of micro organisms in LRN can be broadly classified into two categories: Newtonian and Non-Newtonian. In order to be able to design and control these microrobots, it is necessary to mathematically model them and understand the dynamical behaviour of the system under manipulation of certain parameters. The simplest microrobot capable of locomotion is a mechanism with two rigid links and possessing a single degree of freedom. An early paper(Purcell, 1977) which presented a fundamental theorem related to such a mechanism, discusses the motion characteristics of a scallop-like structure in the LRN regime. From here on, all our future discussions are focused on locomotion in LRN.

2.1.1 Reynolds number

The Reynolds number is an important dimensionless quantity in fluid mechanics used to help predict flow patterns in different fluid flow situations. At LRN flow tends to be dominated by laminar (sheet-like) flow, but at high Reynolds numbers turbulence results from differences in the fluid's speed and direction. It is defined as:

$$Re = \frac{\rho v L}{\mu} = \frac{v L}{\nu} \quad (2.1)$$

where:

- ρ is the density of the fluid
- v is the velocity of the fluid with respect to the object
- L is the characteristic linear dimension of the body
- μ is the dynamic viscosity of the fluid
- ν is the kinematic viscosity of the fluid

Reynolds number, corresponding to the motion of a certain body signifies the ratio of the magnitude of inertial force to the viscous force acting on that body. Hence, a body in the LRN regime experiences a greater magnitude of viscous forces and the inertial terms become insignificant. To be able to get an idea as to what being in the LRN regime implies consider the following example. A micron size body moving in water(which has a kinematic viscosity of about $10^{-2}\text{cm}^2/\text{sec}$, with a velocity of $30\mu\text{m}/\text{sec}$. If the force acting on the body to make it move with the above velocity ceases to act, the approximate distance it moves before coming to rest is $0.1\text{ }\mu\text{m}$ and it takes about $0.6\mu\text{sec}$ to slow down. Hence it can be seen that inertia is completely

irrelevant. In a very LRN regime, motion at that instant is entirely determined by the forces exerted at that moment and by nothing in the past(Purcell, 1977).

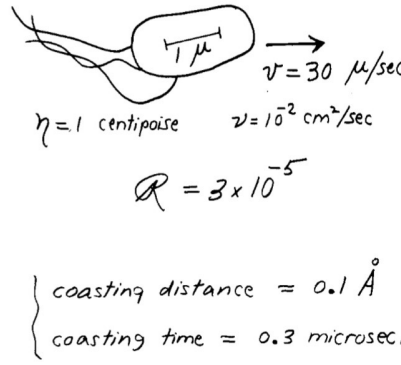


FIGURE 2.1: An example of LRN motion(Purcell, 1977)

2.1.2 Navier-Stokes equation

To explain these observations, consider the Navier Stokes equation given below:

$$\rho \frac{\partial \vec{v}}{\partial t} + \rho(\vec{v} \cdot \nabla) \vec{v} = \eta \nabla^2 \vec{v} - \nabla p \quad (2.2)$$

The inertia terms can be considered to be negligible in the LRN regime. Hence the left hand side goes to zero. Motion in such a regime is known as creeping motion or Stokes flow. Since we are dealing with incompressible flow, creeping flow/ Stokes flow can be represented with the following two equations.

$$\eta \nabla^2 \vec{v} - \nabla p = 0 \quad \nabla \cdot \vec{v} = 0 \quad (2.3)$$

The above equation is linear in space and time independent. Due to its linear nature there are multiple techniques available to obtain exact solutions subject to certain boundary conditions(Leal, 2007). A consequence of this analysis is that in a world of LRN, the response of the fluid to the motion of boundaries is instantaneous. In the limit of very low Re, velocity perturbations diffuse rapidly relative to the rate at which fluid particles are carried along by the flow(Lauga and Powers, 2009). In addition, in the absence of inertia, the swimmer remains perpetually force and torque free(Lauga, 2011).

2.1.3 Scallop theorem

When applied to LRN locomotion, the linearity and time-independence of Stokes equation of motion lead to two important properties.

- The first is rate independence: if a body undergoes surface deformation, the distance travelled by the swimmer between two different surface configurations does not depend on the rate at which the surface deformation occurs but only on its geometry (i.e. the sequence of shapes the swimmer passes through between these two configurations)(Lauga and Powers, 2009).

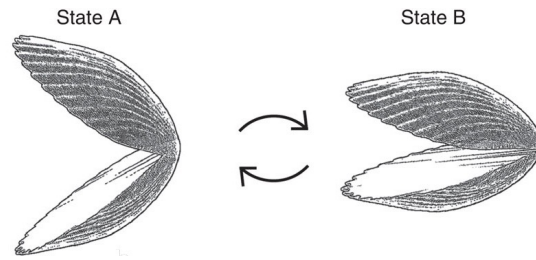


FIGURE 2.2: The open and close configurations of a scallop (Qiu et al., 2014)

- The second important property of swimming without inertia is the so-called scallop theorem: if the sequence of shapes displayed by a swimmer deforming in a time-periodic fashion is identical when viewed after a time-reversal transformation, then the swimmer cannot move on average. Note that the condition is not that the motion be strictly time-reversal invariant, with the same forward and backwards rate, but only that the sequence of shapes is the same when viewed forward or backward in time. This class of surface deformations is termed 'reciprocal deformation'. The scallop theorem introduces a strong geometrical constraint on the type of swimming motion which is effective at LRN (Lauga and Powers, 2009).

Hence, it can be seen that bodies with just one hinge, that possess a single degree of freedom in the configuration space, execute just reciprocal motion, and therefore have no net displacement. However, this phenomenon does not hold in the NN regime, which will be discussed in detail in the sections ahead. In conclusion, to have a finite displacement, the locomotive body needs to possess at least two degrees of freedom. The simplest such structure was proposed by Purcell and is discussed next (Purcell, 1977).

2.1.4 Purcell's swimmer

The simplest model that is capable of executing non reciprocal motion was proposed by E.M. Purcell (Purcell, 1977). This swimmer can be considered as a simplified flagellum made of three slender rods articulated at two hinges, hence, possessing two degrees of freedom. The original form of the Purcell's swimmer has three links moving in a plane, the outer links are actuated through a hinge joint with the central or base link such that the mechanism always performs motion in a plane (Kadam and Banavar, 2016). Figure 2.3 gives us a schematic of the Purcell's swimmer. θ_1 is the angle between the first and the middle link whereas θ_2 is the angle between the middle link and the third link.

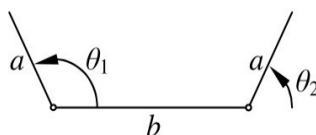


FIGURE 2.3: Schematic illustration of Purcell's swimmer. (Huber, Koehler, and Yang, 2011)

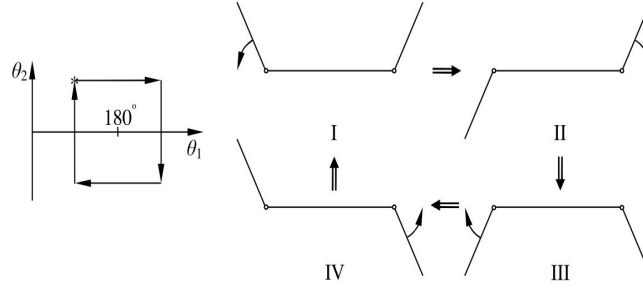


FIGURE 2.4: Schematic illustration of stroking strategy in configuration space (* is the initial state) and stroking sequence in four steps.(Huber, Koehler, and Yang, 2011)

In order to have a finite displacement it follows a loop in the configuration space of its parameters. θ_1 and θ_2 should vary with one another as shown in Figure 2.4. It can be seen that the configuration sequence given in Figure 2.4 is non-reciprocal in time. Due to this non reciprocity, repeated execution of such a motion yields non zero displacement of the swimmer.

Various theories are used to model the fluid bodies in the LRN regime. Most commonly used theories are slender body theory(Johnson and Brokaw, 1979) and resistive force theory(Riley and Lauga, 2017) which will be discussed in the following sections. Prior to that, a brief description of the relevant fluid dynamics of the Newtonian and NN fluids is given in the following section.

2.2 Fluid dynamics

2.2.1 Shear strain rate in a fluid

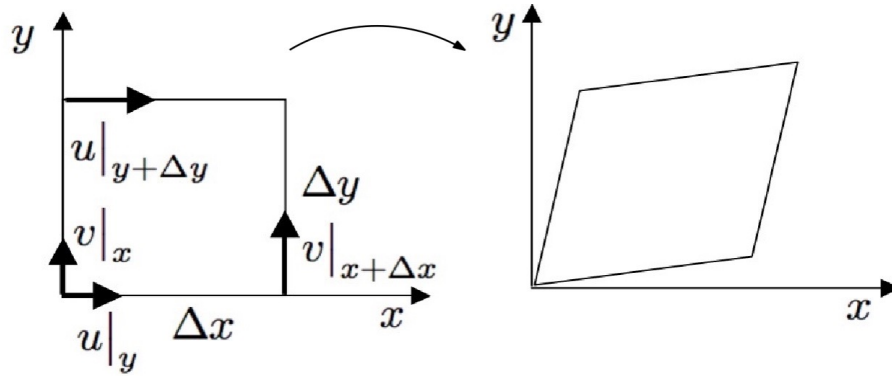


FIGURE 2.5: An infinitesimal volume of fluid undergoing shear (Huber, Koehler, and Yang, 2011)

Consider an infinitesimal fluid element of dimensions Δx , Δy (and also Δz) as shown in 2.5. Let the velocity of the fluid in the x , y and z direction be $u(y, z)$, $v(x, z)$ and $w(x, y)$ respectively. Since the velocity in one direction is a function of spatial coordinates in the other two directions, along with translation, rotation and dilatation the fluid element undergoes shear as well. Since we are interested in shear and the strain rate due to it, we will be considering them in detail. The shear strain rate of a infinitesimal fluid element is the sum of derivatives of the velocities as described below:

$$\dot{\epsilon}_{xy} = \frac{1}{2} \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) \quad (2.4)$$

$$\dot{\epsilon}_{yz} = \frac{1}{2} \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \right) \quad (2.5)$$

$$\dot{\epsilon}_{xz} = \frac{1}{2} \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right) \quad (2.6)$$

The above can be summarised as:

$$\dot{\epsilon}_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \quad (2.7)$$

where, $\dot{\epsilon}_{ij}$ is the shear strain rate in the $i - j$ plane with u_i being the velocity in the i direction. Consider a fluid in motion, the total fluid stress is defined as:

$$\tau_{ij} = -p\delta_{ij} + \hat{\tau}_{ij} \quad (2.8)$$

where p is the thermodynamic pressure and $\hat{\tau}_{ij}$ is the dynamic stress, which, in isotropic fluid is related to the velocities as below

$$\hat{\tau}_{ij} = \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) + \lambda \left(\frac{\partial u_l}{\partial x_l} \right) \quad (2.9)$$

where the fluid properties are:

- μ is the coefficient of dynamic viscosity
- λ is the bulk elasticity, 'second' coefficient of viscosity

For incompressible flow, as a result of continuity, $\frac{\partial u_l}{\partial x_l} = 0$. Therefore, for an incompressible, isotropic, Newtonian fluid the viscous stress $\hat{\tau}_{ij}$ is given as:

$$\hat{\tau}_{ij} = \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \quad (2.10)$$

2.10 describes the behaviour of Newtonian fluids. The coefficient of dynamic viscosity is solely a function of temperature and pressure. But in the case of NN fluids, it is not a linear function of shear rate. The relation between stress and strain rate plays a fundamental role in determining locomotion in fluids. Fluids can be broadly categorised into the following based on the kind of relation between stress and strain rate:

1. Systems for which the value of strain rate at a point within the fluid is determined only by the current value of stress at that point; these substances are variously known as purely viscous, inelastic or time-independent fluids.
2. Systems for which the relation between stress and strain rate shows further dependence on the duration of shearing and kinematic history; these are called time-dependent fluids.
3. Systems which exhibit a blend of viscous fluid behavior and of elastic solid-like behaviour. For instance, this class of materials shows partial elastic recovery, recoil, creep, etc. Accordingly, these are called visco-elastic or elastico-viscous fluids (Chhabra and Richardson, 2008).

We now move on to a brief introduction to both Newtonian and Non-Newtonian fluids.

2.2.2 Newtonian fluids

Most commonly encountered fluids (liquids and gases) such as water, organic solvents, oils, air, steam, nitrogen or rare gases are Newtonian in a wide temperature and pressure range. In contrast to that found for non-Newtonian fluids, the Newtonian viscosity is a physical property which is dependent only on pressure and temperature. Viscosity, as originally was formulated by Newton arises when a layer of fluid is made to move in relation to another layer. Thus, the greater the internal friction, the greater the stress required to promote this movement. Primary difference between Newtonian and NN fluids arise from the differences in their shear stress - strain rate relationship. Hence, it is necessary to understand this relationship in order to categorise the fluids.

2.2.3 Non Newtonian fluids

Since our study concerns NN fluids, we provide a more detailed exposition on this topic. There are many kinds of NN fluids which are briefly described as below (Chhabra and Richardson, 2008).

Time independent fluid behaviour

In simple unidirectional shear, this sub-set of fluids is characterized by the fact that the current value of the rate of shear at a point in the fluid is determined only by the corresponding current value of the shear stress and vice versa. Conversely, one can say that such fluids have no memory of their past history.

$$\dot{\gamma}_{xy} = f(\sigma_{xy}) \quad (2.11)$$

where, $\dot{\gamma}_{xy}$ is the strain rate and σ_{xy} is the stress acting on the fluid body at that instant. Depending upon the form of (2.11), three possibilities exist:

- Shear- thinning or pseudoplastic behavior
- Visco-plastic behavior with or without shear-thinning behavior
- Shear- thickening or dilatant behavior

Shear thinning fluid

Shear thinning fluid is characterized by an apparent viscosity η (defined as $\sigma_{xy} / \dot{\gamma}_{xy}$) which gradually decreases with increasing shear rate. In polymeric systems (melts and solutions), at low shear rates, the apparent viscosity approaches a Newtonian plateau where the viscosity is independent of shear rate (zero shear viscosity, η_0).

$$\lim_{\dot{\gamma}_{xy} \rightarrow 0} \frac{\sigma_{xy}}{\dot{\gamma}_{xy}} = \eta_0 \quad (2.12)$$

Furthermore, only polymer solutions also exhibit a similar plateau at very high shear rates (infinite shear viscosity, η_∞), i.e.,

$$\lim_{\dot{\gamma}_{xy} \rightarrow \infty} \frac{\sigma_{xy}}{\dot{\gamma}_{xy}} = \eta_\infty \quad (2.13)$$

There are various models empirically developed to capture the behaviour of shear thinning fluids, some of which are discussed below.

Viscosity models of shear thinning fluids

- Power law model states that the relationship between shear stress(σ) - shear rate($\dot{\gamma}$) plotted on log-log co-ordinates for a shear-thinning fluid can be approximated by a straight line over an interval of shear rate, i.e.,

$$\sigma = m(\dot{\gamma})^n \quad (2.14)$$

or, in terms of the apparent viscosity,

$$\eta = m(\dot{\gamma})^{n-1} \quad (2.15)$$

This law does not help us capture the asymptotic values of viscosity at very low and very high shear rates.

- Cross viscosity equation is obtained by correcting the power law to obtain the asymptotic limits of viscosity.

$$\frac{\eta - \eta_\infty}{\eta_\infty - \eta_0} = \frac{1}{1 + m(\dot{\gamma})^n} \quad (2.16)$$

In the above two laws, obviously $0 < n < 1$, to have a shear thinning behaviour (viscosity decreasing with increasing shear rate). n is an indication of the extent of shear thinning nature of the material while m is an indication of the consistency of the material.

- Another model which has been found to be more accurate with respect to the experimental result is the Carreau viscosity model (Rao, 2007). The viscosity relationship according to such a model is presented below.

$$\mu_{eff}(\dot{\gamma}) = \mu_{inf} + (\mu_0 - \mu_{inf})(1 + (\lambda\dot{\gamma})^2)^{\frac{n-1}{2}} \quad (2.17)$$

where,

μ_0 is viscosity at shear rate tending to zero

μ_{inf} is viscosity at shear rate tending to infinity

λ is the relaxation time

n is power index

At low shear rate, Carreau fluid behaves as a Newtonian fluid while at higher shear rate as a power law fluid(Rao, 2007).

Viscoplastic fluids

Viscoplastic fluid behavior is characterized by the existence of a threshold stress (called yield stress or apparent yield stress, σ_0) which must be exceeded for the fluid to deform (shear) or flow. Conversely, such a substance will behave like an elastic solid when the externally applied stress is less than the yield stress, σ_0 . Of course, once the magnitude of the external yield stress exceeds the value of σ_0 , the fluid may exhibit Newtonian behaviour (constant value of η) or shear-thinning characteristics, i.e., $\eta(\dot{\gamma})$.

Shear thickening fluids

Shear thickening fluids are similar to pseudoplastic systems in that they show no yield stress, but their apparent viscosity increases with the increasing shear rate and hence the name shear-thickening. The currently available limited information (mostly restricted to simple shear) suggests that it is possible to approximate $\sigma - \dot{\gamma}$ data for these systems also by the power law model, (2.14), with the power-law index n taking on values greater than unity. It is not yet possible to say with confidence whether these materials also display limiting viscosities in the limits of $\dot{\gamma} \rightarrow 0$ and $\dot{\gamma} \rightarrow \infty$. The above time independent behaviour of NN fluids can be visualised with the help of Figure 2.6.

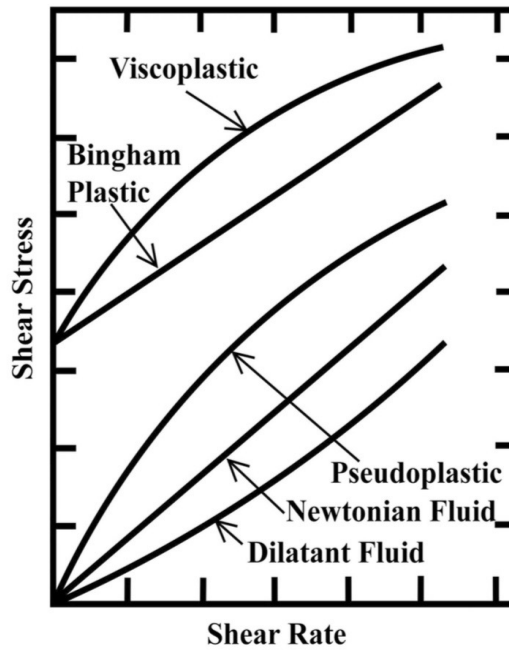


FIGURE 2.6: Shear stress versus shear rate of time independent NN fluids(Chhabra and Richardson, 2008)

Time dependent fluid behaviour

The apparent viscosities of such fluids are not only functions of the applied shear stress σ or the shear rate $\dot{\gamma}$, but also of the duration for which the fluid has been subjected to shearing as well as their previous kinematic history. Depending upon the response of a material to shear over a period of time, it is customary to sub-divide time-dependent fluid behavior into two types:

- Thixotropic fluid behaviour
- Rheopectic (or negative thixotropic) fluid behaviour

Thixotropy

A material is classified as being thixotropic if, when it is sheared at a constant rate, its apparent viscosity $\eta = \frac{\sigma}{\dot{\gamma}}$ (or the value of σ because $\dot{\gamma}$ is constant) decreases with the duration of shearing. If the flow curve of such a fluid is measured in a single experiment in which the value of $\dot{\gamma}$ is steadily increased at a constant rate from zero to some maximum value and then decreased at the same rate, a hysteresis loop of

the form shown schematically in Figure 2.7 is obtained.

Rheopexy

The relatively few fluids which show negative thixotropy, i.e., their apparent viscosity (or the corresponding shear stress) increases with time of shearing are also known as rheopectic fluids. In this case, the hysteresis loop is inverted. (Figure 2.7). Hav-

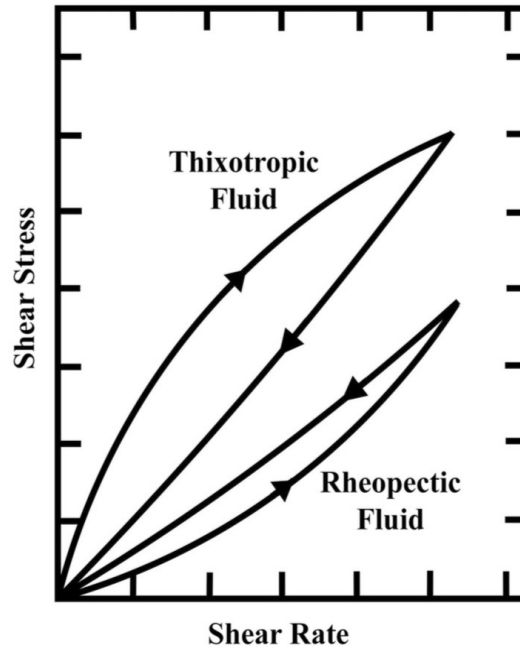


FIGURE 2.7: Shear stress versus shear rate of time dependent NN fluids(Chhabra and Richardson, 2008).

ing understood the differences between Newtonian and NN fluids, there are certain aspects of the NN fluids that can be exploited to overcome the scallop theorem and have non zero displacement with reciprocal motion, which are discussed below.

2.2.4 Locomotion in LRN

To locomote in the NN Regime, the most exploited differences to overcome the reciprocity is discussed below.

1. Complex fluids possess in general rheological properties which are rate dependent. In this type of fluid, and in contrast with the Newtonian case, the rate at which the reciprocal sequence of shapes is being displayed would matter, a result which could be used to design a reciprocal swimmer.
2. The non zero normal stress differences which arise from the stretching by the flow of the microstructure suspended in the complex fluids aid in propulsion
3. Even for small-amplitude motion and linearized dynamics, the simplest evolution equation for the stress in a polymeric fluid contains a memory term in the form of a partial time derivative times a relaxation time. Whether, even in the linear regime, stress relaxation can be taken advantage of for locomotion purposes is an intriguing, but yet unexplored, possibility(Lauga, 2011).

Although the reciprocity fails to hold in NN fluids under certain circumstances, biological swimmers have developed different methods of locomotion to have non zero displacement in Newtonian media. Some of these are discussed below.

- Biological swimmers follow non reciprocal kinematics to overcome the scallop theorem. Most swimming cells locomote by using traveling wave-like deformation of their bodies or appendages.
- Swimming bacteria rotate one or more helical flagella using rotary motors embedded in the cell walls leading to flagella kinematics akin to that of traveling helical waves, and thus propulsion. Basically this motion is characterised by a clear direction of time(Lauga, 2011).

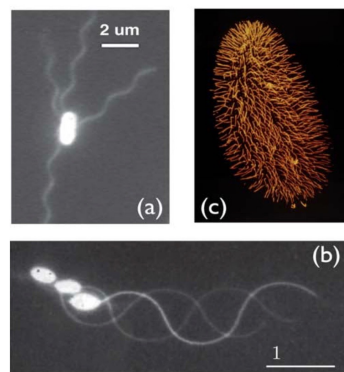


FIGURE 2.8: a) *E. coli* with four helical flagella; (b) Swimming spermatozoon; (c) *Paramecium*'s short cilia(Lauga, 2011)

- Synthetic swimmers may be developed with multiple degrees of freedom to overcome the reciprocity in movement. These degrees of freedom need not necessarily be geometric parameters but physical properties as well, such as, the volume and spontaneous curvature of a lipid vesicle. A reciprocal actuation combined with flexibility or elasticity can lead to kinematics of shape change which are non-reciprocal, and thus to locomotion(Lauga, 2011).

Although a body undergoing reciprocal motion cannot swim, two bodies undergoing reciprocal motion with nontrivial phase differences are able to take advantage of the unsteady hydrodynamic flows they create to undergo nonzero collective and relative dynamics; there is thus no many-scallop theorem.

In order to determine the displacement these bodies undergo, it is necessary to be able to determine the forces acting on them due to the fluid. To model the forces acting on bodies with high aspect ratio (similar to the most commonly prevalent biological specimens) several theories were proposed, of which resistive force theory is the most commonly employed.

Resistive force theory

In order to describe swimming induced by long, slender flagella, resistive force theory was proposed over 60 years ago and has subsequently been improved upon. For a radius of curvature of the flagellar waveform much larger than the diameter of the flagellum, then at leading order in the aspect ratio of the flagellum, the local velocity linearly determines the local force density on the flagellum. The drag can then

be decomposed into the perpendicular and parallel components in this local region. Further refinements include slender-body theory, which provides greater accuracy leading to better qualitative and quantitative approximations. To extend this theory to NN fluids, it is necessary to be able to estimate drag coefficients in such a fluid. Two approaches can be used to calculate the non-Newtonian drag coefficients. The first one is an empirical fit to experimental measurements of sedimenting rods in shear-thinning fluids and thus is directly built from experimental data. The second approach is an ad hoc model based on the Carreau viscosity-shear-rate relationship. Since in shear-thinning fluids the shear viscosity of the fluid is a function of the shear rate, we first need, in this case, a method to estimate accurately the local shear rate around the moving filament. It is done so by approximating the flow as locally Newtonian, allowing us to exploit elementary flow calculations. With this local, instantaneous value of the shear rate, we can then incorporate the shear-thinning nature of the fluid through a correction to the Newtonian drag coefficients and therefore obtain a nonlinear velocity-force relationship. Consider a filament of length $2l$ and radius a as shown in Figure 2.9.

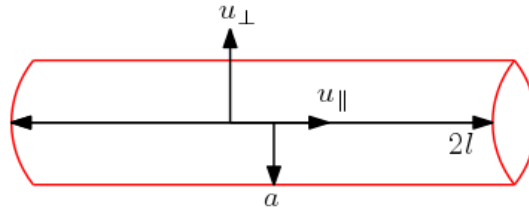


FIGURE 2.9: Straight filament of length $2l$ and cross-sectional radius a may translate in a fluid along its length (velocity $u_{\parallel}$) or perpendicular to it ($u_{\perp}$)(Riley and Lauga, 2017).

The parallel and perpendicular drag coefficients in such a case with a locally Newtonian flow is given by(Riley and Lauga, 2017):

$$\left. \frac{f_{\parallel}}{u_{\parallel}} \right|_{r=a} \equiv b_{\parallel} = \frac{4\pi\mu_0}{\ln\left(\frac{4l^2}{a^2}\right) - 1} \quad (2.18)$$

$$\left. \frac{f_{\perp}}{u_{\perp}} \right|_{r=a} \equiv b_{\perp} = \frac{8\pi\mu_0}{\ln\left(\frac{4l^2}{a^2}\right) + 1} \quad (2.19)$$

Upon further analysis it can be seen that the average shear rate due to both parallel and perpendicular motion is given by:

$$\dot{\gamma}_{avg} = \frac{\sqrt{f_{\perp}^2 + 2f_{\parallel}^2}}{2\sqrt{2}a\pi\mu_0} \quad (2.20)$$

where μ_0 is the Newtonian viscosity of the medium(Riley and Lauga, 2017). In case of NN fluids, μ_0 is no longer a constant and is a function of strain rate. As a result, to propose drag coefficients to use with the Carreau model (or any empirical NN fluid model) we require the knowledge of the shear rates in the fluid near the filament. Consider the Carreau fluid model described for shear thinning fluids as given in 2.17. The viscosity relation for such a model is given by:

$$\mu = \mu_{inf} + (\mu_0 - \mu_{inf})(1 + (\lambda\dot{\gamma})^2)^{\frac{n-1}{2}} \quad (2.21)$$

where,

μ_0 is viscosity at shear rate tending to zero

μ_{inf} is viscosity at shear rate tending to infinity

λ is the relaxation time

n is power index

Since we are operating at the microscale, high shear rates are unlikely to occur, hence we set $\mu_{inf} = 0$, so the model simplifies to

$$\frac{\mu}{\mu_0} = (1 + (\lambda\dot{\gamma})^2)^{\frac{n-1}{2}} \quad (2.22)$$

With the above simplification, we obtain the NN drag coefficients to be:

$$b_{C\perp} = R_C(u_\perp, u_\parallel)b_\perp \quad b_{C\parallel} = R_C(u_\perp, u_\parallel)b_\parallel \quad (2.23)$$

where $b_\parallel$ and $b_\perp$ are given by (2.19) and (2.18) respectively and $R_C(u_\perp, u_\parallel)$ is defined as below:

$$R_C = \left[1 + \left(\frac{\lambda b_\perp v_{\dot{\gamma}}}{2\sqrt{2}a\pi\mu_0} \right)^2 \right]^{\frac{n-1}{2}} \quad (2.24)$$

$v_{\dot{\gamma}}$ is the shear rate velocity defined as:

$$v_{\dot{\gamma}} = \sqrt{v_\perp^2 + 2\frac{b_\parallel^2}{b_\perp^2}v_\parallel^2} \quad (2.25)$$

where $v_\perp$ and $v_\parallel$ are the velocities of the fluid perpendicular and parallel to the slender filament (Riley and Lauga, 2017). However, 2.26 provides us with a linear force density while the scallop is a three dimensional structure. In order to obtain an area force density, Taylor's swimming sheet model is invoked. Taylor's swimming sheet deals with the calculation of viscous forces acting on a sheet immersed in a viscous fluid. For more details on the modelling of Taylor's swimming sheet, refer Lauga and Powers, 2009. The force per unit area on the sheet is the product of the drag coefficient, velocity of the sheet (strain rate in our case) and the wave number of the sheet. Hence the total force density on link 1 can be modelled as below.

$$F_\parallel(s) = b_{C\parallel}(s)u_\parallel(s)k \quad F_\perp(s) = b_{C\perp}(s)u_\perp(s)k, \quad (2.26)$$

The scallop can be assumed to be a rigid segment of a planar sheet whose wavelength is considerably larger than the scallop dimensions. The wave number accounts for the in-extensible planar sheet like structure of the scallop link, keeping the high aspect ratio of the link thickness and length intact. This serves as an extension to the resistive force theory employed in this context. With the above information at hand, we can proceed with the Lagrangian model of the scallop.

Chapter 3

Lagrange model of scallop

This chapter discusses the Lagrangian model of a scallop in detail. Figure 3.1 presents the model of a scallop as two rigid links with a single hinge.

3.1 System definition

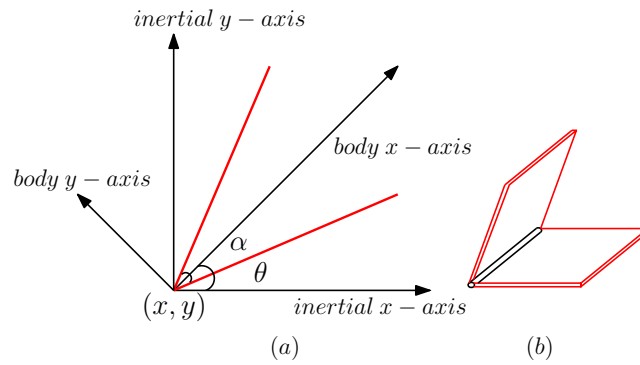


FIGURE 3.1: The schematic of a scallop

The coordinates of the scallop hinge is denoted by (x, y) . The angle between the two links is denoted as α , while the angle between the scallop body x axis and the inertial x axis is given by θ . The length of link 1 is taken as L_1 and the length of link 2 as L_2 . The thickness of the link is $2a$. The dimension of link 1 and link 2 into the plane is L_1 and L_2 respectively. The masses of the two links are denoted by m_1 and m_2 respectively. The coordinates of Link 1 at some distance s_1 along the link is given by,

$$x_1 = x + s_1 \cos\left(\theta + \frac{\alpha}{2}\right) \quad y_1 = y + s_1 \sin\left(\theta + \frac{\alpha}{2}\right) \quad (3.1)$$

where $0 < s_1 < L_1$

Similarly, the coordinates of Link 2 at some distance s_2 is given by,

$$x_2 = x + s_2 \cos\left(\theta - \frac{\alpha}{2}\right) \quad y_2 = y + s_2 \sin\left(\theta - \frac{\alpha}{2}\right) \quad (3.2)$$

where $0 < s_2 < L_2$

On differentiating, we obtain the absolute velocity of the links as,

$$\dot{x}_1 = \dot{x} - s_1 \sin\left(\theta + \frac{\alpha}{2}\right)\left(\dot{\theta} + \frac{\dot{\alpha}}{2}\right) \quad \dot{y}_1 = \dot{y} + s_1 \cos\left(\theta + \frac{\alpha}{2}\right)\left(\dot{\theta} + \frac{\dot{\alpha}}{2}\right) \quad (3.3)$$

$$\dot{x}_2 = \dot{x} - s_2 \sin\left(\theta - \frac{\alpha}{2}\right)\left(\dot{\theta} - \frac{\dot{\alpha}}{2}\right) \quad \dot{y}_2 = \dot{y} + s_2 \cos\left(\theta - \frac{\alpha}{2}\right)\left(\dot{\theta} - \frac{\dot{\alpha}}{2}\right) \quad (3.4)$$

3.2 Total energy of the system

With the above system definition we can compute the kinetic energy of the entire system. The kinetic energy of an element along the links is given by,

$$dT_i = \frac{1}{2} dm(\dot{r}_i \cdot \dot{r}_i) \quad (3.5)$$

where,

$$\dot{r}_i = \dot{x}_i \hat{i} + \dot{y}_i \hat{j}$$

Total kinetic energy is given by:

$$T = \int_0^{L_1} dT_1 + \int_0^{L_2} dT_2 \quad (3.6)$$

Since, there are no conservative forces acting on the system, the system possesses no potential energy. The total energy of the system is its kinetic energy, which can be represented in the form of a matrix as shown below.

$$T = \begin{bmatrix} \dot{x} & \dot{y} & \dot{\theta} & \dot{\alpha} \end{bmatrix} \begin{bmatrix} \frac{m_1+m_2}{2} & 0 & -\frac{m_2 L_2}{2} \sin\left(\theta - \frac{\alpha}{2}\right) & \frac{m_2 L_2}{4} \sin\left(\theta - \frac{\alpha}{2}\right) \\ 0 & \frac{m_1+m_2}{2} & \frac{m_2 L_2}{2} \cos\left(\theta - \frac{\alpha}{2}\right) & -\frac{m_2 L_2}{4} \cos\left(\theta - \frac{\alpha}{2}\right) \\ -\frac{m_1 L_1}{2} \sin\left(\theta + \frac{\alpha}{2}\right) & \frac{m_1 L_1}{2} \cos\left(\theta + \frac{\alpha}{2}\right) & \frac{m_1 L_1^2 + m_2 L_2^2}{6} & \frac{m_1 L_1^2}{6} \\ -\frac{m_1 L_1}{4} \sin\left(\theta + \frac{\alpha}{2}\right) & \frac{m_1 L_1}{4} \cos\left(\theta + \frac{\alpha}{2}\right) & -\frac{m_2 L_2^2}{6} & \frac{m_1 L_1^2 + m_2 L_2^2}{24} \end{bmatrix} \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \\ \dot{\alpha} \end{bmatrix}$$

3.3 Forces on the links

The force density acting on the links is taken as the product of the wave number (of the hypothetical sheet the scallop is a segment of) and drag coefficient times the strain rate, which in this case is the velocity of the link. The value of this drag coefficient varies according to the strain rate -as it's a NN fluid- and is a medium property. Since the opening and closing velocities of the scallop are different, we have different values for this coefficient, in the direction parallel to and perpendicular to the link, during the two strokes. The longitudinal drag coefficient is indicated by $b_{C_{\parallel}}$ and the lateral drag coefficient is by $b_{C_{\perp}}$ and are given by (2.23). The velocity perpendicular and parallel to the links at any point, in this case for link 1 is given by:

$$v_{\parallel} = \dot{x}_1 \cos\left(\theta + \frac{\alpha}{2}\right) + \dot{y}_1 \sin\left(\theta + \frac{\alpha}{2}\right) \quad v_{\perp} = \dot{y}_1 \cos\left(\theta + \frac{\alpha}{2}\right) - \dot{x}_1 \sin\left(\theta + \frac{\alpha}{2}\right) \quad (3.7)$$

with $\dot{x}_1$ and $\dot{y}_1$ as defined in (3.3).

The force density on link 1 is given by:

$$F_{\parallel}(s) = b_{C_{\parallel}}(s) v_{\parallel}(s) k \quad F_{\perp}(s) = b_{C_{\perp}}(s) v_{\perp}(s) k \quad (3.8)$$

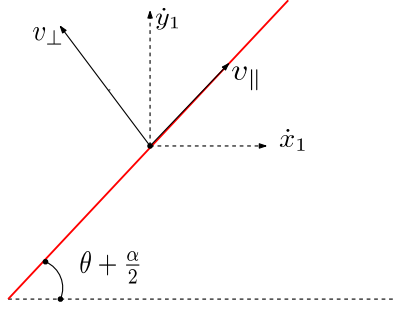


FIGURE 3.2: Velocity at the centre of link 1

$b_{C_{\parallel}}, b_{C_{\perp}}, v_{\parallel}, v_{\perp}$ all vary with distance along the link, the force density also varies as a function of position along the link.

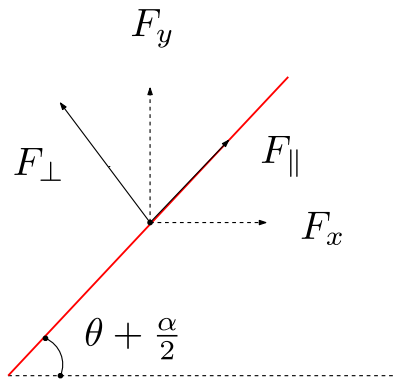


FIGURE 3.3: Force acting on link 1

Figure 3.3 shows the equivalent force density distribution on the link. The force densities acting on link 1 are found to be,

$$F_{x1}(s) = F_{\parallel}(s) \cos\left(\theta + \frac{\alpha}{2}\right) - F_{\perp}(s) \sin\left(\theta + \frac{\alpha}{2}\right) \quad (3.9)$$

$$F_{y1}(s) = F_{\parallel}(s) \sin\left(\theta + \frac{\alpha}{2}\right) + F_{\perp}(s) \cos\left(\theta + \frac{\alpha}{2}\right) \quad (3.10)$$

Similarly the force density can be computed for the second link with the only change being the angle between the second link and the horizontal axis of the spatial frame. To compute the complete force acting on link 1 it is necessary to perform element wise integration of the force density along the link length. The total force f_{x1} in the x direction is:

$$f_{x1} = L_1 * \int_0^{L_1} F_{x1}(s) ds \quad (3.11)$$

The total force f_{y1} in the y direction is:

$$f_{y1} = L_1 * \int_0^{L_1} F_{y1}(s) ds \quad (3.12)$$

Similarly we can obtain the forces f_{x2} and f_{y2} acting on link 2.

With the above computations of energy, velocities and forces, Euler Lagrange equations can now be derived.

3.4 Euler-Lagrange equations

The Lagrange's Equations of Motion are of the following form,

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_j}\right) - \frac{\partial T}{\partial q_j} = Q_j \quad \forall j = 1, 2, \dots, n \quad (3.13)$$

where,

$$Q_j = \sum_{i=1}^N F_i \frac{\partial r_i}{\partial q_j}$$

where r_i is the position vector of the i th link, q_j represents a generalised coordinate and T represents the total kinetic energy of the system.

The F_i term here refers to the force acting on the link i .

This system comprises of four generalised coordinates, namely, x , y , θ and α .

Hence, this, along with the system definition provides us with four equations of motion which define the system.

Equation of motion corresponding to x

Presented below is the equation corresponding to the generalised coordinate ' x '.

$$\begin{aligned} f_{x1} + f_{x2} = & \ddot{x}(m_1 + m_2) - \frac{\dot{\theta}}{2}[m_1 L_1 \cos\left(\theta + \frac{\alpha}{2}\right)(\dot{\theta} + \frac{\dot{\alpha}}{2}) + m_2 L_2 \cos\left(\theta - \frac{\alpha}{2}\right)(\dot{\theta} - \frac{\dot{\alpha}}{2})] \\ & - \frac{\ddot{\theta}}{2}[m_1 L_1 \sin\left(\theta + \frac{\alpha}{2}\right) + m_2 L_2 \sin\left(\theta - \frac{\alpha}{2}\right)] - \frac{\dot{\alpha}}{4}[m_1 L_1 \cos\left(\theta + \frac{\alpha}{2}\right)(\dot{\theta} + \frac{\dot{\alpha}}{2}) \\ & - m_2 L_2 \cos\left(\theta - \frac{\alpha}{2}\right)(\dot{\theta} - \frac{\dot{\alpha}}{2})] - \frac{\ddot{\alpha}}{4}[m_1 L_1 \sin\left(\theta + \frac{\alpha}{2}\right) - m_2 L_2 \sin\left(\theta - \frac{\alpha}{2}\right)] \end{aligned} \quad (3.14)$$

where f_{x1} and f_{x2} are as described in subsection 4.3.

Equation of motion corresponding to y

Presented below is the equation corresponding to the generalised coordinate ' y '.

$$\begin{aligned} f_{y1} + f_{y2} = & \ddot{y}(m_1 + m_2) - \frac{\dot{\theta}}{2}[m_1 L_1 \sin\left(\theta + \frac{\alpha}{2}\right)(\dot{\theta} + \frac{\dot{\alpha}}{2}) + m_2 L_2 \sin\left(\theta - \frac{\alpha}{2}\right)(\dot{\theta} - \frac{\dot{\alpha}}{2})] \\ & + \frac{\ddot{\theta}}{2}[m_1 L_1 \cos\left(\theta + \frac{\alpha}{2}\right) + m_2 L_2 \cos\left(\theta - \frac{\alpha}{2}\right)] - \frac{\dot{\alpha}}{4}[m_1 L_1 \sin\left(\theta + \frac{\alpha}{2}\right)(\dot{\theta} + \frac{\dot{\alpha}}{2}) \\ & - m_2 L_2 \sin\left(\theta - \frac{\alpha}{2}\right)(\dot{\theta} - \frac{\dot{\alpha}}{2})] + \frac{\ddot{\alpha}}{4}[m_1 L_1 \cos\left(\theta + \frac{\alpha}{2}\right) - m_2 L_2 \cos\left(\theta - \frac{\alpha}{2}\right)] \end{aligned} \quad (3.15)$$

where f_{y1} and f_{y2} are as described in subsection 4.3.

Equation of motion corresponding to θ

The torque due to the fluid forces is due to the perpendicular drag force as shown below:

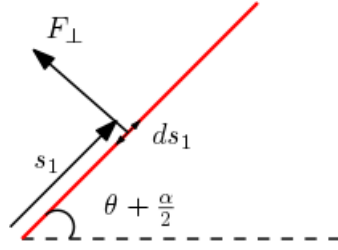


FIGURE 3.4: Elemental calculation of torque

External torque is applied for the actuation (opening and closing) of the links.

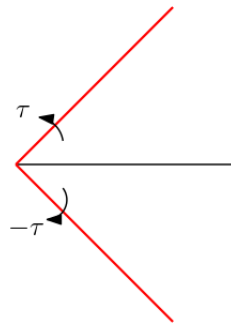


FIGURE 3.5: External torque applied for actuation

The torque acting on an element on link 1 is given by:

$$d\tau_1 = s_1 * F_{\perp}(s) * ds_1 * L_1 \quad (3.16)$$

The total torque acting on link 1 is given by:

$$\tau_1 = \int_0^{L_1} d\tau_1$$

Similarly τ_2 can be computed with the angle of link 2 with the spatial x axis being $(\theta - \frac{\alpha}{2})$, where τ_1, τ_2 are the torques acting on links 1 and 2 due to the fluid viscosity and τ is the external torque applied to actuate the scallop.

Presented below is the equation corresponding to the generalised coordinate ' θ '.

$$\begin{aligned}
\tau_1 + \tau_2 = & + \frac{\dot{\theta}\dot{y}}{2}[m_1L_1\sin\left(\theta + \frac{\alpha}{2}\right) + m_2L_2\sin\left(\theta - \frac{\alpha}{2}\right)] + \frac{\dot{\theta}\dot{x}}{2}[m_1L_1\cos\left(\theta + \frac{\alpha}{2}\right) \\
& + m_2L_2\cos\left(\theta - \frac{\alpha}{2}\right)] + \frac{\dot{\alpha}\dot{y}}{4}[m_1L_1\sin\left(\theta + \frac{\alpha}{2}\right) - m_2L_2\sin\left(\theta - \frac{\alpha}{2}\right)] \\
& + \frac{\dot{\alpha}\dot{x}}{4}[m_1L_1\cos\left(\theta + \frac{\alpha}{2}\right) - m_2L_2\cos\left(\theta - \frac{\alpha}{2}\right)] + \frac{\ddot{\theta}}{3}[m_1L_1^2 + m_2L_2^2] \\
& + \frac{\ddot{\alpha}}{6}[m_1L_1^2 - m_2L_2^2] + \frac{\ddot{y}}{2}[m_1L_1\cos\left(\theta + \frac{\alpha}{2}\right) + m_2L_2\cos\left(\theta - \frac{\alpha}{2}\right)] \\
& - \frac{\ddot{x}}{2}[m_1L_1\sin\left(\theta + \frac{\alpha}{2}\right) + m_2L_2\sin\left(\theta - \frac{\alpha}{2}\right)] - \frac{\dot{y}}{2}[m_1L_1\sin\left(\theta + \frac{\alpha}{2}\right)(\dot{\theta} + \frac{\dot{\alpha}}{2}) \\
& + m_2L_2\sin\left(\theta - \frac{\alpha}{2}\right)(\dot{\theta} - \frac{\dot{\alpha}}{2})] - \frac{\dot{x}}{2}[m_1L_1\cos\left(\theta + \frac{\alpha}{2}\right)(\dot{\theta} + \frac{\dot{\alpha}}{2}) \\
& + m_2L_2\cos\left(\theta - \frac{\alpha}{2}\right)(\dot{\theta} - \frac{\dot{\alpha}}{2})]
\end{aligned} \tag{3.17}$$

Equation of motion corresponding to α

Presented below is the equation corresponding to the generalised coordinate ' α '.

$$\begin{aligned}
\frac{\tau_1 - \tau_2}{2} + \tau = & + \frac{\dot{\theta}\dot{y}}{4}[m_1L_1\sin\left(\theta + \frac{\alpha}{2}\right) - m_2L_2\sin\left(\theta - \frac{\alpha}{2}\right)] + \frac{\dot{\theta}\dot{x}}{4}[m_1L_1\cos\left(\theta + \frac{\alpha}{2}\right) \\
& - m_2L_2\cos\left(\theta - \frac{\alpha}{2}\right)] + \frac{\dot{\alpha}\dot{y}}{8}[m_1L_1\sin\left(\theta + \frac{\alpha}{2}\right) + m_2L_2\sin\left(\theta - \frac{\alpha}{2}\right)] \\
& + \frac{\dot{\alpha}\dot{x}}{8}[m_1L_1\cos\left(\theta + \frac{\alpha}{2}\right) + m_2L_2\cos\left(\theta - \frac{\alpha}{2}\right)] + \frac{\ddot{\alpha}}{12}[m_1L_1^2 + m_2L_2^2] \\
& + \frac{\ddot{\theta}}{6}[m_1L_1^2 - m_2L_2^2] + \frac{\ddot{y}}{4}[m_1L_1\cos\left(\theta + \frac{\alpha}{2}\right) - m_2L_2\cos\left(\theta - \frac{\alpha}{2}\right)] \\
& - \frac{\ddot{x}}{4}[m_1L_1\sin\left(\theta + \frac{\alpha}{2}\right) - m_2L_2\sin\left(\theta - \frac{\alpha}{2}\right)] - \frac{\dot{y}}{4}[m_1L_1\sin\left(\theta + \frac{\alpha}{2}\right)(\dot{\theta} + \frac{\dot{\alpha}}{2}) \\
& - m_2L_2\sin\left(\theta - \frac{\alpha}{2}\right)(\dot{\theta} - \frac{\dot{\alpha}}{2})] - \frac{\dot{x}}{4}[m_1L_1\cos\left(\theta + \frac{\alpha}{2}\right)(\dot{\theta} \\
& + \frac{\dot{\alpha}}{2}) - m_2L_2\cos\left(\theta - \frac{\alpha}{2}\right)(\dot{\theta} - \frac{\dot{\alpha}}{2})]
\end{aligned} \tag{3.18}$$

where τ_1 , τ_2 , τ are as defined in the previous equation of motion corresponding to θ .

3.5 Results and discussions

For the purpose of simulating (3.14) to (3.18) on MATLAB, the physical characteristics of the scallop were taken as denoted in the following table.

TABLE 3.1: Microscallop Properties

Scallop Property	Value
link length	0.8mm
link breadth	0.8mm
link thickness	0.1mm
mass of single link	$61.76 \times 10^{-9} \text{kg}$

Although all the analysis in the previous sections have been with respect to shear thinning fluids, we perform simulations for both shear thinning and thickening fluids. Although the theory holds for shear thinning alone, we find that the simulation results hold true for shear thickening fluids as well.

3.5.1 Shear thinning fluids

The shear thinning fluid was taken to possess the following Carreau coefficients:

TABLE 3.2: Shear Thinning Fluid Properties

Fluid Property	Value
Zero strain rate viscosity	0.377 Pa-s
Shear thinning coefficient (n)	0.3
Relaxation time	0.512s

The initial angle between the links is $\alpha = 0.5618$ radians and the initial angle between the scallop axis and the horizontal axis is $\theta = 0.523$ radians. The scallop can be assumed to be a part of a thin sheet, whose wave number is $1m^{-1}$ Upon simulating the four equations, the following displacement trends were observed. In all but the last case, the torque applied has a time period of 4 seconds.

1. **Symmetric actuation:** Under symmetric actuation, with similar opening and closing speeds there is negligible net displacement as expected. Since the resistive force depends on the strain rate, the force acting during the opening and closing of the scallop is identical if the opening and closing strain rates are similar, thereby leading to no net displacement. Upon a symmetric actuation of external torque as shown in Figure 3.6, the behaviour of the scallop is as depicted in Figure 3.7 and Figure 3.8.

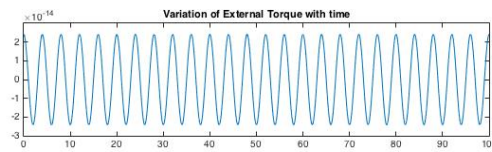


FIGURE 3.6: Variation of the external torque acting on the scallop

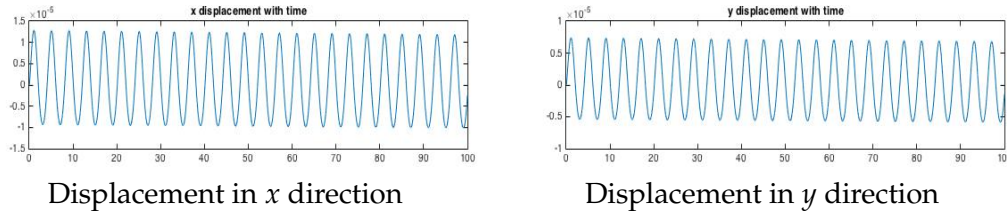


FIGURE 3.7: Displacement in metres against time in seconds

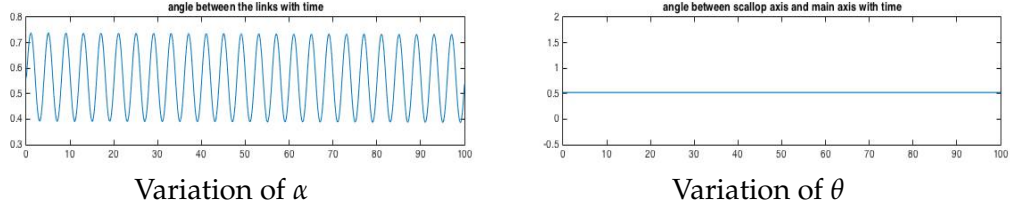


FIGURE 3.8: Displacement in radians against time in seconds

2. **Asymmetric actuation:** Upon asymmetric actuation of the scallop as indicated by Figure 3.9, with slow opening and fast closing of the links and the fluid properties as defined in Table 3.3, the displacement trends are as shown in Figure 3.10 and 3.21. Due to differential strain rates, the resistive forces during the two strokes are of different magnitudes, leading to a non zero displacement at the end of every cycle.

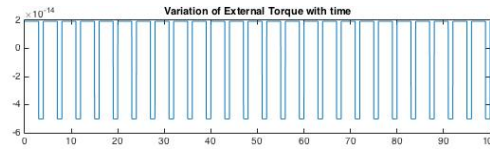


FIGURE 3.9: Variation of the external torque acting on the scallop

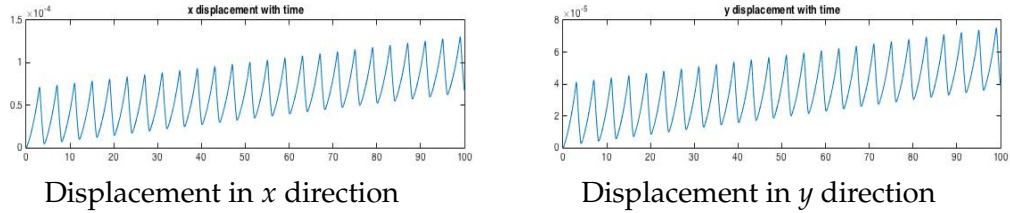


FIGURE 3.10: Displacement in metres against time in seconds

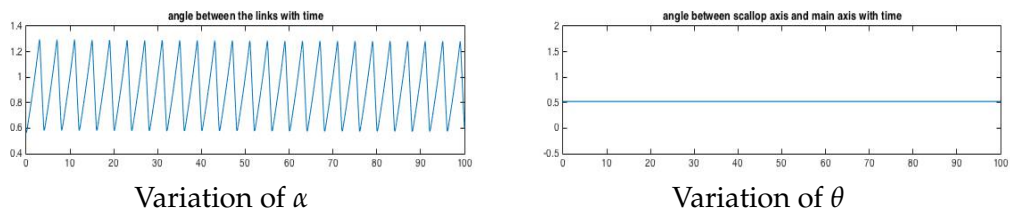


FIGURE 3.11: Displacement in radians against time in seconds

3. **Higher zero shear rate viscosity:** Upon increasing the zero shear rate viscosity to $\mu_0 = 0.6$, the drag coefficients get scaled to a higher value and consequently the resistive force increases accordingly leading to a decrease in the overall displacement to around $60\mu m$ in 100s as can be observed in Figures 3.12 and 3.23.

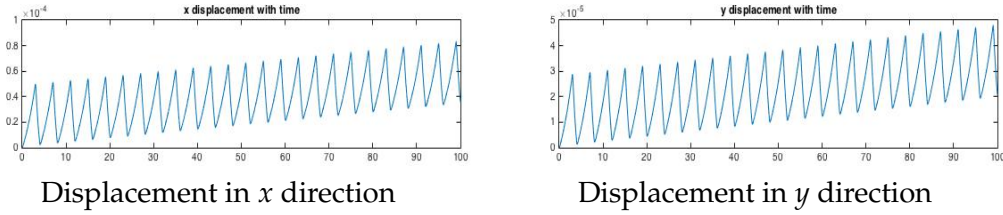


FIGURE 3.12: Displacement in metres against time in seconds

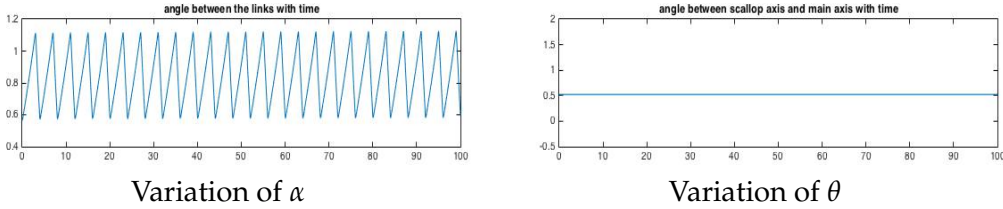


FIGURE 3.13: Displacement in radians against time in seconds

4. **Reduced shear thinning coefficient:** Upon decreasing the shear thinning coefficient to $n = 0.2$, the resistive force decreases (as the shear rate has a greater effect on the viscosity with decreasing shear thinning coefficient) accordingly leading to an increase in the overall displacement to around $200\mu\text{m}$ in 100s as can be observed in Figures 3.26 and 3.27.

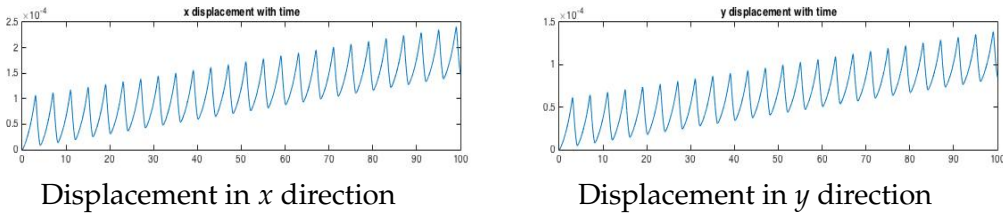


FIGURE 3.14: Displacement in metres against time in seconds

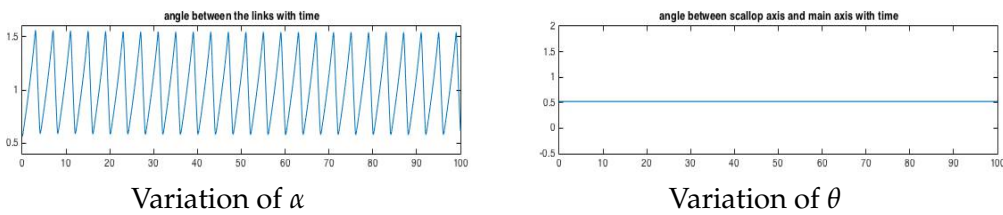


FIGURE 3.15: Displacement in radians against time in seconds

5. **Increased time period:** When the time period of the actuation torque is increased to 6s (from 4s in the previous simulations) as shown in Figure 3.16, the opening and closing torques act for a longer period of time, leading to the links reaching higher velocities. This results in an overall reduction in the resistive forces acting on the link, thereby leading to a greater net displacement. Only the time period of the torque has been altered, while the magnitude is almost a constant. This behaviour is depicted in Figures 3.17 and 3.18, with the fluid properties as described in Table 3.3. There is a significant increase in the overall displacement as when compared to asymmetric actuation of time period 4(case 2).

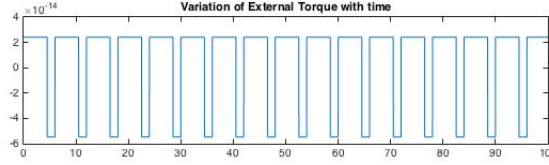


FIGURE 3.16: Variation of the external torque acting on the scallop

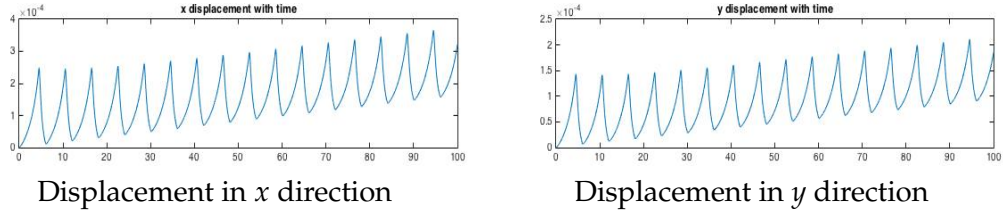


FIGURE 3.17: Displacement in metres against time in seconds

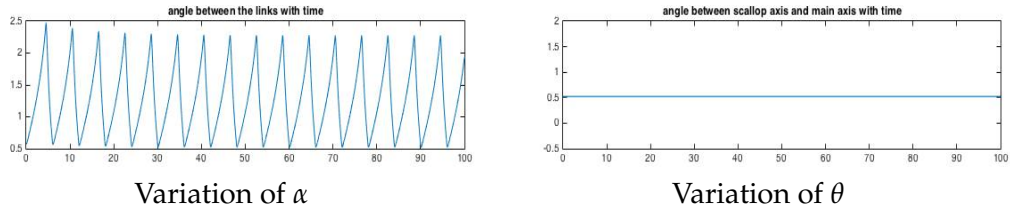


FIGURE 3.18: Displacement in radians against time in seconds

From the above figures we can observe that the velocity of the scallop along θ is nearly equal to $1 - 2\mu\text{m/s}$ depending on the actuation.

3.5.2 Shear thickening fluids

The time period of the torque actuation is 4 seconds. The shear thickening fluid was chosen to possess the following Carreau coefficients (as used in experiments in Qiu et al., 2014:

TABLE 3.3: Shear Thickening Fluid Properties

Fluid Property	Value
Zero strain rate viscosity	0.8 Pa-s
Shear thickening coefficient (n)	2.34
Relaxation time	0.67s

The following situations were simulated:

1. **Asymmetric actuation:** Similar to the actuation in the shear thinning case, we subject the microscallop to slow opening and fast closing. Due to the shear force-strain rate relationship being opposite to what is in the case of a shear thinning fluid, we obtain a trend negative to the corresponding shear thinning cases. That is, the force acting on the scallop while it is opening, is lesser in magnitude than when it is closing. This is because the closing speeds are higher, and due to the shear thickening nature of the fluid, the resistive force

experienced increases with increasing shear rate. This is in opposition to the shear thinning case, where the forces acting while the scallop is opening is higher in magnitude than while it is closing. Since the forces acting on the scallop are of opposite nature in the shear thickening case in comparison to the thinning fluid, the scallop experiences net displacement in a direction opposite to that of the shear thinning one.

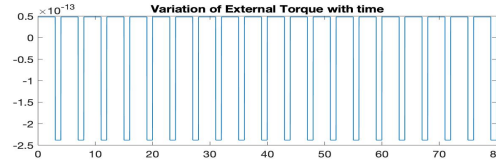


FIGURE 3.19: Variation of the external torque acting on the scallop

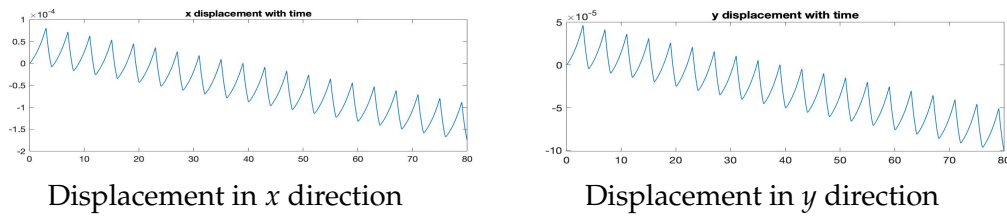


FIGURE 3.20: Displacement in metres against time in seconds

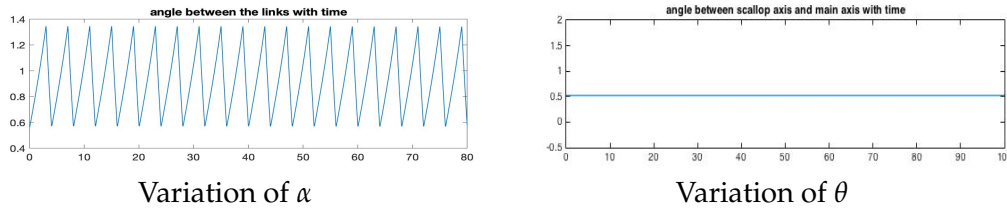


FIGURE 3.21: Displacement in radians against time in seconds

2. **Higher zero shear rate viscosity:** Upon increasing the zero shear rate viscosity from $\mu_0 = 0.8$ to $\mu_0 = 1$, under asymmetric actuation as before, the resistive force increases accordingly leading to a decrease by approximately $70\mu\text{m}$ in the overall displacement in a time period of 80s as observed in the following plots. The reasoning is similar to the shear thinning case, that is, since the drag coefficient directly depends on the zero shear viscosity, increasing it will lead to an increase in drag coefficient magnitude, further resulting in an increase in the resistive force experienced by the scallop, leading to reduced displacements.

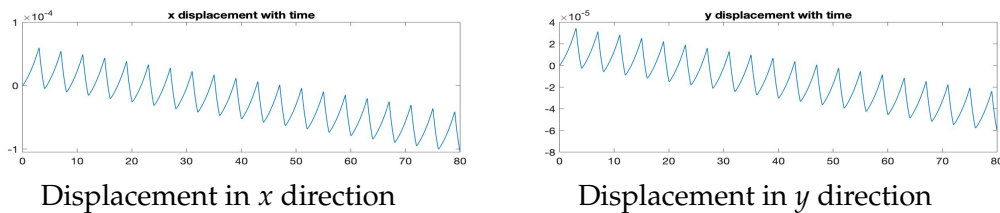


FIGURE 3.22: Displacement in metres against time in seconds

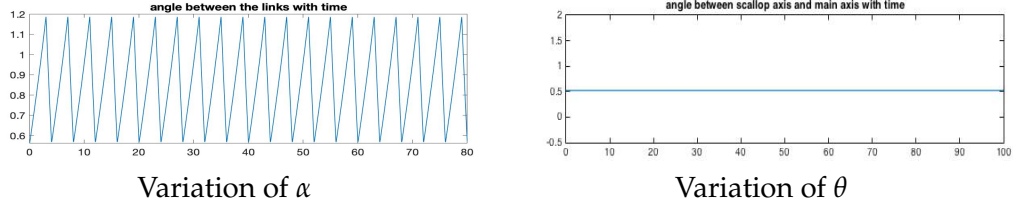


FIGURE 3.23: Displacement in radians against time in seconds

3. **Increase shear thickening coefficient:** Upon increasing the shear thickening coefficient from $n = 2.34$ to $n = 2.75$, under similar asymmetric actuation, the resistive force increases (as the shear rate has a greater effect on the viscosity with increasing shear thickening coefficient) accordingly leading to an decrease by approximately $50\mu m$ in the overall displacement in a time period of 80s as observed in the following plots. The reasoning is similar to the shear thinning case, however, here as n is increasing, the effect of increasing velocities on increasing drag coefficients are higher. Therefore, we observe an increase in the resistive force and a drop in displacement.

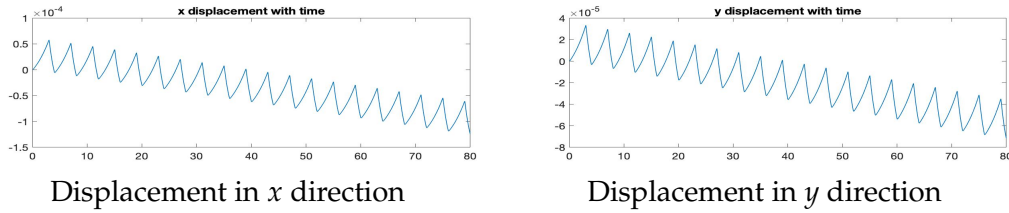


FIGURE 3.24: Displacement in metres against time in seconds

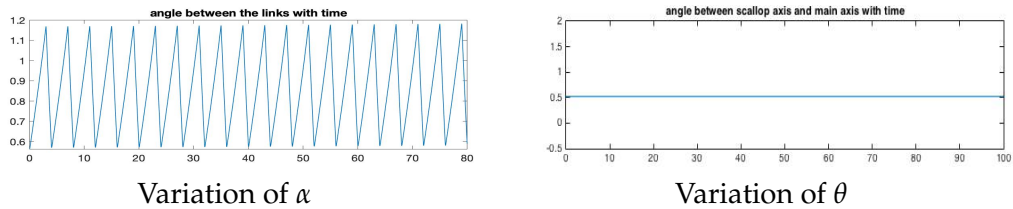


FIGURE 3.25: Displacement in radians against time in seconds

4. **Reverse actuation scheme - fast open, slow close:** Upon reversing the actuation scheme, the direction of the displacement of the scallop also reverses. When the scallop is subject to fast open and slow close, the resistive force magnitudes in both legs of the cycle get reversed. That is, the resistive force experienced while opening is higher than while closing, as opening speeds are higher than closing speeds. Therefore the displacement of the scallop in a shear thickening fluid is qualitatively similar to the displacement of the scallop in a shear thinning fluid with a reversed actuation scheme.

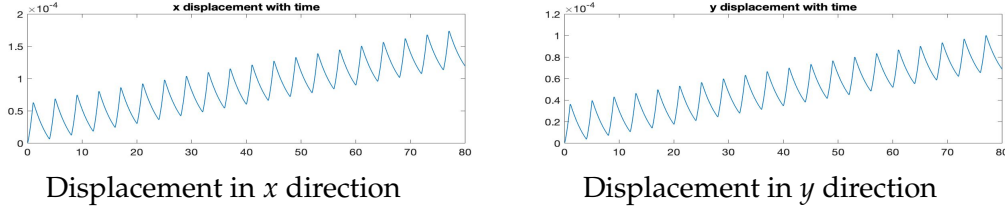


FIGURE 3.26: Displacement in metres against time in seconds

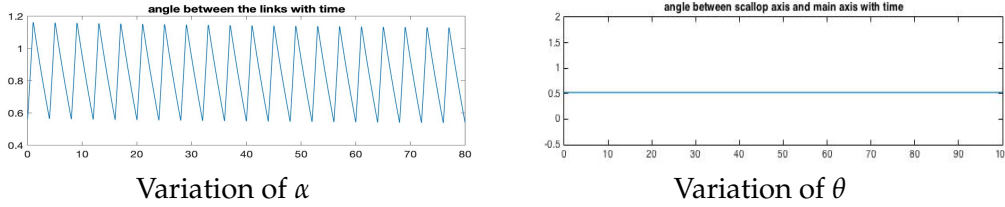


FIGURE 3.27: Displacement in radians against time in seconds

From the above figures we can observe that the velocity of the scallop along θ is nearly equal to $2 - 3 \mu\text{m/s}$ depending on the actuation.

Since we were keen on validating our theoretical approach and numerical experiments with realistic observations and data, we compared our results with that obtained by an experimental group in Qiu et al., 2014. For a microscallop of similar dimensions and fluid properties, our technique, which is model based, predicts almost similar displacements as observed through experiments. The asymmetric actuation of the scallop in experiments conducted by Qiu et al., 2014 is orchestrated by periodically varying, externally controlled magnetic fields.

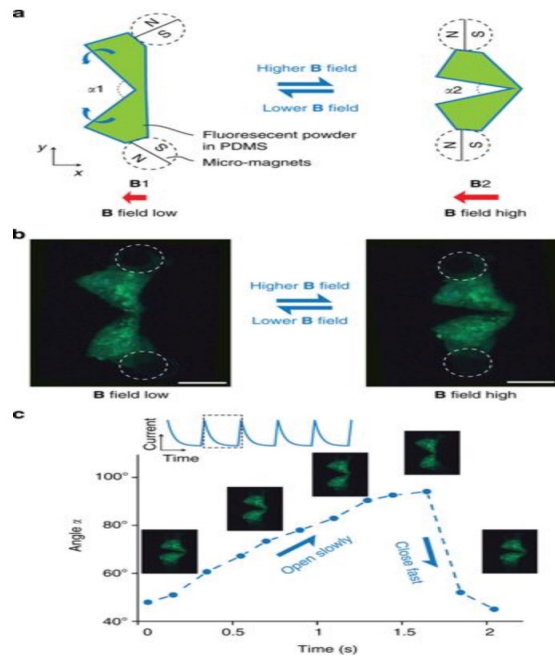


FIGURE 3.28: a) The schematic of a scallop being subject to periodically varying magnetic field. b) Top view of microscallop under ultra-violet illumination (white circles indicate the position of the magnets). c) Variation of angle between the links over one cycle of actuation (Qiu et al., 2014).

The simulation results presented at the beginning of this chapter follows a similar actuation scheme as followed in 3.28, that is slow opening (3s) and fast closing (1s). The scallop material and dimensions for the purpose of simulations are assumed to be the same as those in these experiments. Fluid properties are chosen along similar lines for accurate corroboration.

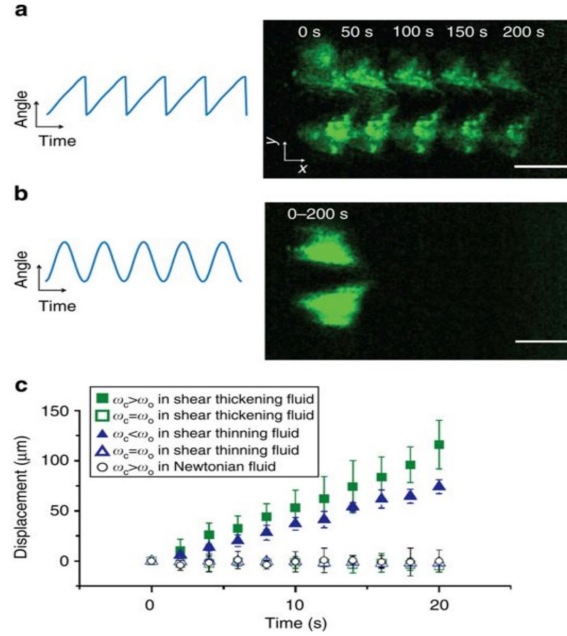


FIGURE 3.29: a) Non zero scallop displacement upon asymmetric actuation (slow open and fast close). b) No net displacement upon symmetric actuation of scallop. c) Scallop displacements under different actuation schemes in shear thickening, shear thinning and Newtonian fluids (Qiu et al., 2014).

3.29 illustrates displacement of scallop in variation actuation schemes, which are enlisted in the above simulations. It can be observed that the velocity of the scallop in experiments and the numerical simulations are of the same order of magnitude. These experimental results serve in validating our mathematical model of the microscallop in NN fluids.

Reduction in model complexity

Since our larger objective was to develop control algorithms, we decided to analyze the system for certain control-theoretic properties. In order to proceed with this analysis it is necessary to reduce the complexity, in order to effectively utilise the available computational tools. Two main assumptions have been incorporated, which have been verified to not affect the simulation results and at the same time, provide us with simpler dynamics to work with. The assumptions are as follows:

1. Discretisation of the force and torque integral: The force and torque integrals which appear in the Euler Lagrange equations (3.14-3.18) are significantly complicated and by the virtue of it, lack a closed analytical form. Since the dimension of the scallop links in these simulations are $800\mu m$, discretising the links into twenty individual segments and using the lower sum to approximate the integral, yields accurate results as in the case without this approximation. The

number of discretised segments may vary with link length. This approximation greatly reduced the computational input required for further analysis.

2. The drag coefficients in case of the NN fluids is a correction factor(3.19) times the Newtonian drag coefficient. Assuming the shear rate velocity is small enough in magnitude, we can use a binomial approximation of the following kind:

$$R_C = \left[1 + \frac{(n^2 - 1)}{4} \left(\frac{\lambda b_{\perp} v_{\dot{\gamma}}}{2\sqrt{2}a\pi\mu_0} \right)^2 \right] \quad (3.19)$$

The exponent nature of the correction factor was a major point of complexity, which has been alleviated through the above approximation, which once again doesn't hamper with the expected simulation trends.

With these simplifications, we move onto the next section.

3.6 Microscallop accessibility

In this section we explore a notion called accessibility at certain configurations of the microscallop. To do so we introduce certain standard control and notation terminology.

A control-affine system $\Sigma = (M, \mathcal{C} = \{f_0, f_1, \dots, f_m\}, U)$ has associated with it the differential equations

$$\dot{\gamma}(t) = f_0(\gamma(t)) + \sum_{a=1}^m u^a(t) f_a(\gamma(t)) \quad (3.20)$$

A controlled trajectory of Σ is a pair (γ, u) where $u : I \rightarrow U$ is locally integrable and where $\gamma : I \rightarrow M$ is the corresponding locally absolutely continuous curve satisfying 3.20. The set of controlled trajectories for Σ is denoted by $\text{Ctraj}(\Sigma)$ and the set of controlled trajectories for Σ defined on $[0, T]$ is denoted by $\text{Ctraj}(\Sigma, T)$. Various states that can be reached by trajectories of such a control-affine system can be defined now. For $x_0 \in M$, set of states reachable in time exactly T is defined by,

$$R_{\Sigma}(x_0, T) = \{\gamma(T) | (\gamma, u) \in \text{Ctraj}(\Sigma, T), \gamma(0) = x_0\} \quad (3.21)$$

The set of states reachable in time at most T is given by $R_{\Sigma}(x_0, T) = \cup_{t \in (0, T)} R_{\Sigma}(x_0, t)$. The set of states reachable from x_0 we denote by $R_{\Sigma}(x_0)$. Note that $x_0 \in R_{\Sigma}(x_0, T)$ for any $T > 0$.

Definition 3.6.1 Let $\Sigma = (M, \mathcal{C} = \{f_0, f_1, \dots, f_m\}, U)$ be a C^{∞} control affine system and let $x_0 \in M$, then Σ is **accessible** from x_0 if there exists $T > 0$ such that $\text{int}R_{\Sigma}(x_0, T) \neq \emptyset$ for $t \in [0, T]$ (Bullo and Lewis, 2004).

The following theorem provides a necessary and sufficient condition to check for a system's accessibility.

Theorem 3.6.1 Let $\Sigma = (M, \mathcal{C}, U)$ be an analytic control affine system with U almost proper. Then Σ is accessible from x_0 if and only if $\text{Lie}^{(\infty)}(\mathcal{C}_{x_0}) = T_{x_0}M$ (Bullo And Lewis, 2004).

With the above tools, the accessibility of the microscallop system in a NN fluid can be studied. For the purpose of our simulations we have assumed the torques acting on the two links to be of equal magnitude and opposite in sign. But under the

assumption that the two links can be subject to two different torques, gives rise to two control vector fields. Hence, the four Euler Lagrange Equations(3.14-3.18) of the scallop system can be decoupled and written as:

$$\dot{x}(t) = f_0(x(t)) + u^1 f_1(x(t)) + u^2 f_2(x(t)) \quad (3.22)$$

where $x(t)$ is a curve in an eight dimensional state space M (corresponding to the four generalised coordinates and their velocities), u^1 and u^2 are the scalar torques acting on the two links and $\{u^1, u^2\} \in U$, where U is a subset in $\mathbb{R}^2$. The vector field f_0 is the drift vector field, describing the dynamics of the system in the absence of controls, and the vector fields f_1, f_2 are the input vector fields or control vector fields, through which the control inputs influence the system evolution.

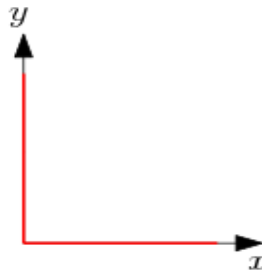


FIGURE 3.30: Scallop configuration: $x = 0, y = 0, \theta = 0.79rad$ and $\alpha = 1.57rad$

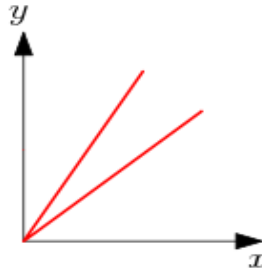


FIGURE 3.31: Scallop configuration: $x = 0, y = 0, \theta = 0.92rad$ and $\alpha = 0.26rad$

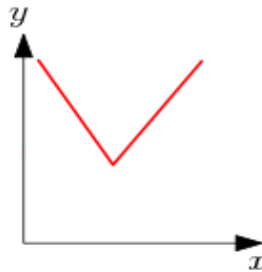


FIGURE 3.32: Scallop configuration: $x = 1m, y = 1m, \theta = 1.57rad$ and $\alpha = 1.05rad$

All the above configurations are assumed to be at rest (zero rotational and translational velocities). The Lie brackets computed for the purpose of accessibility evaluation were

$$[f_0, f_1], [f_0, f_2], [f_0, [f_0, f_1]], [f_0, [f_0, f_2]], [f_1, [f_0, f_1]], [f_1, [f_0, [f_0, f_1]]]$$

These brackets along with f_1 and f_2 were found to span $T_{x_0}M$. All the above symbolic computations were conducted on Mathematica. For values of these brackets for one of these configurations, please refer to [B](#). Hence, it can be concluded from above that Σ is accessible from these locations and possible other configurations as well.

The above model's predictions have been corroborated by priorly conducted experiments (Qiu et al., [2014](#)). The fundamental premise of these results is the breakdown of the scallop theorem in NN fluids. However, another possible method to overcome this theorem in Newtonian fluids is by having multiple scallops with non trivial phase differences. Since most of the microrobots work in coordination with one another rather than in solo, this particular aspect is studied and modelled in detail in the upcoming chapters.

Chapter 4

Coordinated scallop motion - mathematical modelling

Most applications of microrobots require their coordinated control to achieve any task. The previous chapters discuss the dynamics of a microscallop undergoing reciprocal motion in NN fluids. Since the viscosity of such fluids depend on the shear rate the fluid is being subjected to, reciprocal deformations with asymmetric rate of actuation do lead to net displacement (as the force experienced in the forward and return strokes are different, if the rate of the strokes are different). However, the scallop theorem is relevant only in case of a single scallop being suspended in a quiescent Newtonian fluid.

However, when there are multiple scallops, as long as they have non trivial phase differences, reciprocal motion in a Newtonian fluid will cause them to have certain displacement (Lauga and Bartolo, 2008, Koiller, Ehlers, and Montgomery, 1996).

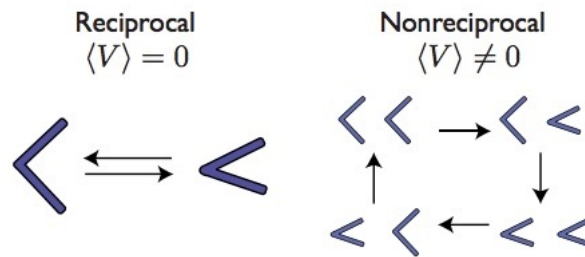


FIGURE 4.1: More than one scallop with non-trivial phase differences undergoing reciprocal motion do undergo non-zero finite displacement (Lauga and Bartolo, 2008.)

Each scallop is influenced by the hydrodynamic disturbances caused by the motion of the other scallops. These hydrodynamic influences breakdown the reciprocity of the system (Lauga and Bartolo, 2008).

In order to understand these hydrodynamics which bind a scallop to its neighbours, it is necessary to be able to model them. To reduce the complexity of these interactions, the far field description of swimmer locomotion is studied. These far field descriptions can be utilised to study the effect of one swimmer onto another.

4.1 Stokeslet and Stokes dipole

In the LRN regime, if a point force, denoted by $f\vec{e}\delta(x)$ (where $\delta(x)$ is the Dirac delta function), is acting at the origin in a quiescent fluid, the Navier Stokes equation can be represented as:

$$\nabla p = \mu \nabla^2 \vec{v} + f\vec{e}\delta \quad (4.1)$$

$$\nabla \cdot \vec{v} = 0 \quad (4.2)$$

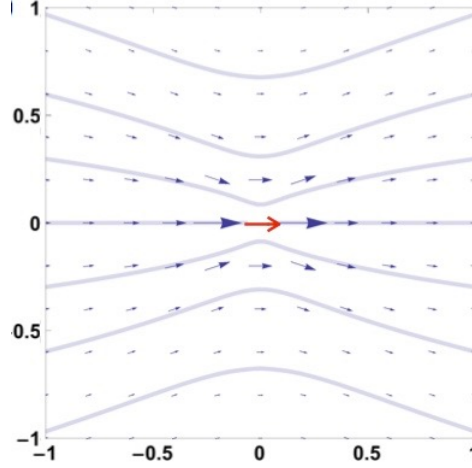


FIGURE 4.2: Velocity field due to a point force acting at the origin. Red vector represents the force, while the blue corresponds to the Stokeslet (Pak and Lauga, 2016).

4.1 and 4.2 can be solved to obtain the velocity field of the fluid and is given by:

$$\vec{v}(\vec{x}) = f\vec{G}(\vec{x};\vec{e}) = \frac{f}{8\pi\mu} \left[\frac{\vec{e}}{r} + \frac{(\vec{e} \cdot \vec{x})\vec{x}}{r^3} \right] \quad (4.3)$$

where $r = |\vec{x}|$ is the distance from the point of application of the force. This fundamental singular solution in viscous flows is called a Stokeslet. Since it decays as $\frac{1}{r}$, the velocity field is long ranged.

As the scallop undergoes reciprocal motion, the force generated by the two links are opposite to one another, thereby the net force exerted by the scallop on the surrounding fluid is zero. However, since these forces are separated by some distance (which is smaller than the link dimensions), they act as a dipole, giving rise to a non zero velocity field as detailed below.

Stokes dipole can be perceived as the limit at which the distance between two Stokeslets of equal magnitude and opposite directions becomes infinitesimally small and the strength is adjusted to give a non zero finite flow field. Consider two stokeslets separated by a distance d . Let the first stokeslet be represented by $\vec{v}^+(\vec{x})$, with a strength of $f\vec{e}$ acting at a distance $\frac{d}{2}$ from the origin along the direction $\vec{d}$ (Lauga and Bartolo, 2008).

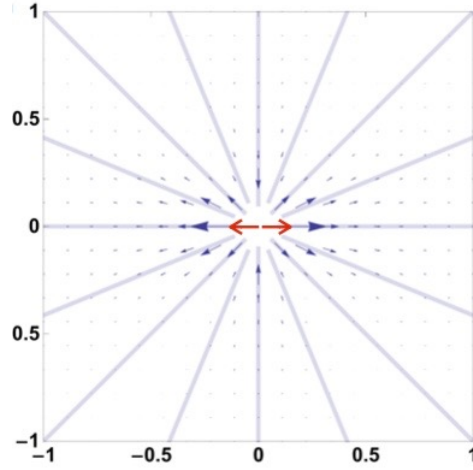


FIGURE 4.3: Velocity field (Stokes dipole) due to a force dipole. The force dipole is represented in red. The blue corresponds to the resulting velocity field (Pak and Lauga, 2016).

The stokeslet can be obtained through Taylor's expansion about the origin as following:

$$\vec{v}^+(\vec{x}) = f\vec{G}(\vec{x} - d\frac{\vec{d}}{2}; \vec{e}) = f\left[\vec{G}(\vec{x}; \vec{e}) - \frac{d}{2}\vec{d} \cdot \nabla \vec{G}(\vec{x}; \vec{e}) + \dots\right] \quad (4.4)$$

Similarly, for another Stokeslet, $\vec{v}^-(\vec{x})$, of opposite strength $-f\vec{e}$ acting in the opposite direction and at a distance of $-d\frac{\vec{d}}{2}$, through Taylor's expansion about the origin we get the following expression:

$$\begin{aligned} \vec{v}^-(\vec{x}) &= f\vec{G}(\vec{x} + d\frac{\vec{d}}{2}; -\vec{e}) = f\left[\vec{G}(\vec{x}; -\vec{e}) + \frac{d}{2}\vec{d} \cdot \nabla \vec{G}(\vec{x}; -\vec{e}) + \dots\right] \\ &= f\left[-\vec{G}(\vec{x}; \vec{e}) - \frac{d}{2}\vec{d} \cdot \nabla \vec{G}(\vec{x}; \vec{e}) + \dots\right] \end{aligned} \quad (4.5)$$

To obtain the overall flow field these two Stokeslets need to be overlapped.

$$\vec{v}(\vec{x}) = \vec{v}^+(\vec{x}) + \vec{v}^-(\vec{x}) = -df\vec{d} \cdot \nabla \vec{G}(\vec{x}; \vec{e}) + \dots \quad (4.6)$$

$\vec{G}(\vec{x}; \vec{e})$ is given in 4.3.

$$\vec{G}(\vec{x}; \vec{e}) = \vec{G}(\vec{x}; \vec{e}) = \frac{f}{8\pi\mu} \left[\frac{\vec{e}}{r} + \frac{(\vec{e} \cdot \vec{x})\vec{x}}{r^3} \right] \quad (4.7)$$

By taking one derivative of an $\vec{e}$ directed Stokeslet along the $\vec{d}$ direction, will provide us with the expression for $\vec{d} \cdot \nabla \vec{G}(\vec{x}; \vec{e})$.

$$\begin{aligned} \vec{G}_D(\vec{x}; \vec{d}, \vec{e}) &= -\vec{d} \cdot \nabla \vec{G}(\vec{x}; \vec{e}) = \frac{1}{8\pi\mu} \left[\frac{(\vec{e}\vec{d} - \vec{d}\vec{e}) \cdot \vec{x}}{r^3} + \left[\frac{3}{2}(\vec{e}\vec{d} + \vec{d}\vec{e}) - (\vec{e} \cdot \vec{d})\vec{I} \right] : \frac{\vec{x}\vec{x}\vec{x}}{r^5} \right] \\ &= \frac{1}{8\pi\mu} \left[\frac{(\vec{d} \times \vec{e}) \times \vec{x}}{r^3} - \left[\frac{(\vec{e} \cdot \vec{d})\vec{x}}{r^3} - 3\frac{(\vec{e} \cdot \vec{x})(\vec{d} \cdot \vec{x})\vec{x}}{r^5} \right] \right] \end{aligned} \quad (4.8)$$

The straining motion of the fluid is represented by a stresslet(4.9) , which corresponds to the flow due to the symmetric part of the Stokes dipole tensor.

$$\vec{S}(\vec{x}; \vec{d}, \vec{e}) = \frac{1}{8\pi\mu} \left[-\frac{(\vec{e} \cdot \vec{d})\vec{x}}{r^3} + 3\frac{(\vec{e} \cdot \vec{x})(\vec{d} \cdot \vec{x})\vec{x}}{r^5} \right] \quad (4.9)$$

The flow due to a point torque, known as a rotlet, is the flow due to the antisymmetric term of the Stokes dipole.

$$\vec{R}(\vec{x}; \vec{d}, \vec{e}) = \frac{(\vec{d} \times \vec{e}) \times \vec{x}}{8\pi\mu r^3} \quad (4.10)$$

The leading order contribution to the velocity field of such an arrangement of forces is known as a Stokes dipole. As long as the distance d is sufficiently small, higher order terms become negligible leaving with the Stokes dipole term alone(Pak and Lauga, 2016).

With the above introduction to a few results regarding the velocity field, in the Stokes regime, we move onto incorporate these results into our two scallop modelling.

4.2 Coordinated scallop system properties

In 3 we studied the scallop system definition in a NN fluid. We derived the dynamics of such a scallop. However, this section presents the kinematic model of a two-scallop system. Since we are dealing with 2 dimensional scallop systems, the following figure provides us with the coordinates completely defining each individual scallop.

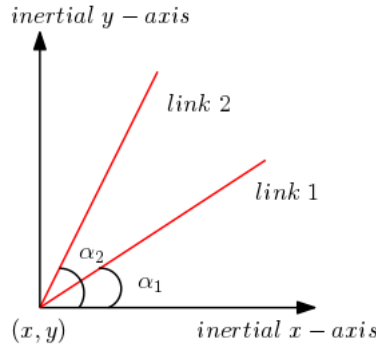


FIGURE 4.4: Schematic of a microscallop

The coordinates of the scallop hinge are denoted by (x, y) . The angle between the first link and the inertial x axis is α_1 and the angle between the second link and the inertial x axis is α_2 . The scallop links are assumed to be of the same length, that is l . The links thickness is denoted as a .

Velocity of link 1, parallel and perpendicular to the link, at any distance s along the length of the link is given by:

$$v_{\parallel}^1 = \dot{x} \cos \alpha_1 + \dot{y} \sin \alpha_1 \quad (4.11)$$

$$v_{\perp}^1 = \dot{y} \cos \alpha_1 + \dot{x} \sin \alpha_1 + \dot{\alpha}_1 s \quad (4.12)$$

As a consequence of swimming at LRN, we observe the following results (Hatton and Choset, 2013):

- The drag forces acting on the links are linear functions of their velocities.
- Since the viscous forces heavily dominate the inertial forces in the LRN regime, the sum of forces acting on the scallop go to zero. The scallop can traverse with velocities which lead to zero force acting on the system. Any fluctuations in these velocities will be damped out by the viscous forces of the fluid.

The linear drag forces density on such a link in a Newtonian fluid is given by 2.18 and 2.19:

$$\left. \frac{f_{\parallel}}{u_{\parallel}} \right|_{r=a} \equiv b_{\parallel} = \frac{4\pi\mu_0}{\ln\left(\frac{4l^2}{a^2}\right) - 1} \quad (4.13)$$

$$\left. \frac{f_{\perp}}{u_{\perp}} \right|_{r=a} \equiv b_{\perp} = \frac{8\pi\mu_0}{\ln\left(\frac{4l^2}{a^2}\right) + 1} \quad (4.14)$$

From the above equations it is evident that if we assume the drag coefficient in the direction parallel to the link to be k , then the drag coefficient in the direction perpendicular to the link is approximately $2k$. Then from 4.11 and 4.12, the forces acting on link 1 in the direction parallel and perpendicular to the links are given by:

$$f_{\parallel}^1 = \int_0^l v_{\parallel}^1 k ds \quad (4.15)$$

$$f_{\parallel}^1 = kl [\dot{x} \cos \alpha_1 + \dot{y} \sin \alpha_1] \quad (4.16)$$

$$f_{\perp}^1 = \int_0^l v_{\perp}^1 (2k) ds \quad (4.17)$$

$$f_{\perp}^1 = 2kl \left[\dot{y} \cos \alpha_1 - \dot{x} \sin \alpha_1 + \frac{\dot{\alpha}_1 l}{2} \right] \quad (4.18)$$

Similarly for link 2:

$$f_{\parallel}^2 = kl [\dot{x} \cos \alpha_2 + \dot{y} \sin \alpha_2] \quad (4.19)$$

$$f_{\perp}^2 = 2kl \left[\dot{y} \cos \alpha_2 - \dot{x} \sin \alpha_2 + \frac{\dot{\alpha}_2 l}{2} \right] \quad (4.20)$$

The force acting on the scallop is given by:

$$f_x^1 = f_{\parallel}^1 \cos \alpha_1 - f_{\perp}^1 \sin \alpha_1 \quad f_y^1 = f_{\parallel}^1 \sin \alpha_1 + f_{\perp}^1 \cos \alpha_1 \quad (4.21)$$

$$\begin{aligned} f_x^1 &= kl [\dot{x} \cos^2 \alpha_1 + \dot{y} \sin \alpha_1 \cos \alpha_1] - 2kl \left[\dot{y} \cos \alpha_1 \sin \alpha_1 - \dot{x} \sin^2 \alpha_1 + \frac{\dot{\alpha}_1 l \sin \alpha_1}{2} \right] \\ &= \dot{x} [kl \cos^2 \alpha_1 + 2kl \sin^2 \alpha_1] - \dot{y} kl \sin \alpha_1 \cos \alpha_1 - \frac{2kl^2 \dot{\alpha}_1 \sin \alpha_1}{2} \\ &= \dot{x}(kl)(1 + \sin^2 \alpha_1) - \dot{y} kl \sin \alpha_1 \cos \alpha_1 - kl^2 \dot{\alpha}_1 \sin \alpha_1 \end{aligned}$$

$$\begin{aligned} f_y^1 &= kl [\dot{x} \cos \alpha_1 \sin \alpha_1 + \dot{y} \sin^2 \alpha_1] + 2kl \left[\dot{y} \cos^2 \alpha_1 - \dot{x} \sin \alpha_1 \cos \alpha_1 + \frac{\dot{\alpha}_1 l \cos \alpha_1}{2} \right] \\ &= \dot{x}(-kl)(\sin \alpha_1 \cos \alpha_1) + \dot{y}(kl)(1 + \cos^2 \alpha_1) + (kl)(l\dot{\alpha}_1 \cos \alpha_1) \end{aligned}$$

Similarly for link 2 we have:

$$f_x^2 = \dot{x}(kl)(1 + \sin^2 \alpha_2) - \dot{y}kl \sin \alpha_2 \cos \alpha_2 - kl^2 \ddot{\alpha}_2 \sin \alpha_2$$

$$f_y^2 = \dot{x}(-kl)(\sin \alpha_2 \cos \alpha_2) + \dot{y}(kl)(1 + \cos^2 \alpha_2) + (kl)(l\ddot{\alpha}_2 \cos \alpha_2)$$

The net force acting on the scallop has to be zero. This leads to the following results:

$$f_x^1 + f_x^2 = 0 \quad f_y^1 + f_y^2 = 0 \quad (4.22)$$

$$\dot{x}(2 + \sin^2 \alpha_1 + \sin^2 \alpha_2) - \dot{y}(\sin \alpha_1 \cos \alpha_1 + \sin \alpha_2 \cos \alpha_2) = (l\ddot{\alpha}_1 \sin \alpha_1 + l\ddot{\alpha}_2 \sin \alpha_2) \quad (4.23)$$

$$-\dot{x}(\sin \alpha_1 \cos \alpha_1 + \sin \alpha_2 \cos \alpha_2) + \dot{y}(2 + \cos^2 \alpha_1 + \cos^2 \alpha_2) = -(l\ddot{\alpha}_1 \cos \alpha_1 + l\ddot{\alpha}_2 \cos \alpha_2) \quad (4.24)$$

The above yields the velocity of the scallop hinge to be expressed in terms of its rotational link velocities as follows:

$$\begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = \begin{bmatrix} K_{11} & K_{12} \\ K_{21} & K_{22} \end{bmatrix} \begin{bmatrix} \dot{\alpha}_1 \\ \dot{\alpha}_2 \end{bmatrix} \quad (4.25)$$

where,

$$K_{11} = \frac{(2l \sin \alpha_1 + l \cos^2 \alpha_2 \sin \alpha_1 - l \cos \alpha_1 \cos \alpha_2 \sin \alpha_2)}{G(\alpha_1, \alpha_2)}$$

$$K_{12} = \frac{(2l \sin \alpha_2 + l \cos^2 \alpha_1 \sin \alpha_2 - l \cos \alpha_2 \cos \alpha_1 \sin \alpha_1)}{G(\alpha_1, \alpha_2)}$$

$$K_{21} = \frac{(-2l \cos \alpha_1 - l \cos \alpha_1 \sin^2 \alpha_2 + l \cos \alpha_2 \sin \alpha_1 \sin \alpha_2)}{G(\alpha_1, \alpha_2)}$$

$$K_{22} = \frac{(-2l \cos \alpha_2 - l \cos \alpha_2 \sin^2 \alpha_1 + l \cos \alpha_1 \sin \alpha_2 \sin \alpha_1)}{G(\alpha_1, \alpha_2)}$$

and $G(\alpha_1, \alpha_2)$ is given by:

$$G(\alpha_1, \alpha_2) = (\cos^2 \alpha_1 \sin^2 \alpha_2 + \cos^2 \alpha_2 \sin^2 \alpha_1 - 2 \cos \alpha_1 \cos \alpha_2 \sin \alpha_1 \sin \alpha_2 + 8)$$

A consequence of zero force on the scallop results is the following relation:

$$\begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = K(\alpha_1, \alpha_2) \begin{bmatrix} \dot{\alpha}_1 \\ \dot{\alpha}_2 \end{bmatrix} \quad (4.26)$$

The above zero force kinematics holds for each individual scallop. However, along with the above, each scallop influences the velocity of the other based on their hydrodynamic interactions, which are modelled in detail as below.

4.2.1 Hydrodynamic coupling

Consider a two microscallop system - microscallop A and B as shown in 4.15. By the virtue of the motion of links of A, a velocity field is generated by the force dipole that A generates. Then, by superposition, the total velocity of scallop B is the sum of its velocity due to angular motion of its links and the Stokes dipole of A. Velocity of A is calculated along similar lines.

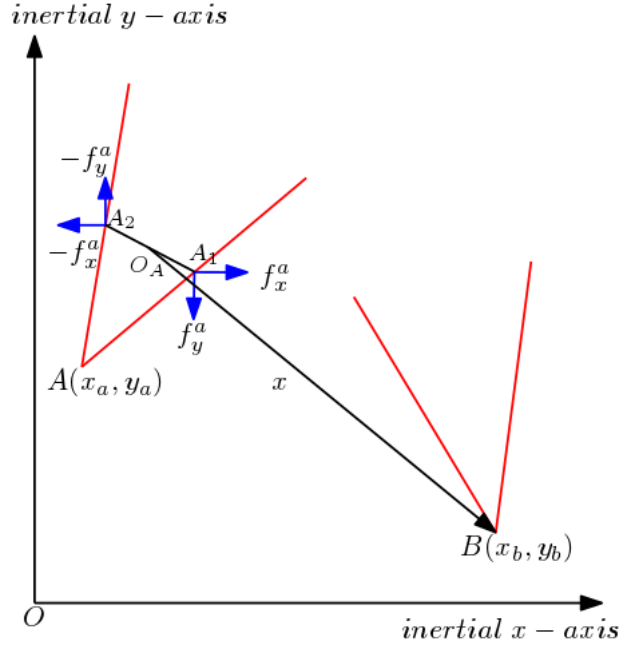


FIGURE 4.5: Schematic of influence of motion of scallop A on B.

$$A1 : ((x_a + l \cos \alpha_1), (y_a + l \sin \alpha_1))$$

$$A2 : ((x_a + l \cos \alpha_2), (y_a + l \sin \alpha_2))$$

The force dipole is oriented along direction d given by,

$$\vec{d} = A2 - A1 = l(\cos \alpha_2 - \cos \alpha_1)\hat{i} + l(\sin \alpha_2 - \sin \alpha_1)\hat{j}$$

Computing the Stokes dipole corresponding to f_x^A yields:

$$\hat{e} = \hat{i}$$

The location of the center of the force dipole of microscallop A is:

$$O_A = \left(x_a + \frac{l}{2}(\cos \alpha_1 + \cos \alpha_2) \right), \left(y_a + \frac{l}{2}(\sin \alpha_1 + \sin \alpha_2) \right)$$

Distance of scallop B from the center of this force dipole is given by:

$$x_{BA} = B - O_A$$

$$x_{BA} = \left(x_b - x_a - \frac{l}{2}(\cos \alpha_1 + \cos \alpha_2) \right), \left(y_b - y_a - \frac{l}{2}(\sin \alpha_1 + \sin \alpha_2) \right)$$

$$r = |x_{BA}|$$

Velocity field at B due to f_x^A dipole is computed according to 4.6:

$$\vec{V}(x) = \frac{df_x^A}{8\pi\mu} \vec{h}(\alpha_{1A}, \alpha_{2A}, x_a, x_b, y_a, y_b)$$

The magnitude of the force f_x^A in the above expression is expressed as:

$$f_x^A = \dot{x}_a(kl)(1 + \sin^2 \alpha_{1A}) - \dot{y}_a(kl)(\sin \alpha_{1A} \cos \alpha_{1A}) - kl(l\dot{\alpha}_{1A} \sin \alpha_{1A}),$$

where $\dot{x}_a$ and $\dot{y}_a$ of scallop A can be expressed in terms of its link angular velocities due to the net zero force property as illustrated in 4.25:

$$\dot{x}_a = K_{11}^A \dot{\alpha}_{1A} + K_{12}^A \dot{\alpha}_{2A}$$

$$\dot{y}_a = K_{21}^A \dot{\alpha}_{1A} + K_{22}^A \dot{\alpha}_{2A}$$

$$\begin{aligned} f_x^A &= (kl)(1 + \sin^2 \alpha_{1A})(K_{11}^A \dot{\alpha}_{1A} + K_{12}^A \dot{\alpha}_{2A}) - \\ &\quad (kl)(\sin \alpha_{1A} \cos \alpha_{1A})(K_{21}^A \dot{\alpha}_{1A} + K_{22}^A \dot{\alpha}_{2A}) - (kl)(l\dot{\alpha}_{1A} \sin \alpha_{1A}) \\ &= \dot{\alpha}_{1A} [(kl)K_{11}^A(1 + \sin^2 \alpha_{1A}) - (kl)(K_{21}^A)(\sin \alpha_{1A} \cos \alpha_{1A}) - kl^2 \sin \alpha_{1A}] \\ &\quad + \dot{\alpha}_{2A} [(kl)K_{12}^A(1 + \sin^2 \alpha_{1A}) - (kl)(K_{22}^A)(\sin \alpha_{1A} \cos \alpha_{1A})] \\ &= [G_{11} \quad G_{12}] \begin{bmatrix} \dot{\alpha}_{1A} \\ \dot{\alpha}_{2A} \end{bmatrix} \end{aligned} \quad (4.27)$$

This yields:

$$\vec{v}(x) = \begin{bmatrix} V_x^{BA_x} \\ V_y^{BA_x} \end{bmatrix} = \frac{d}{8\pi\mu} [G_{11}^A \quad G_{12}^A] \begin{bmatrix} \dot{\alpha}_{1A} \\ \dot{\alpha}_{2A} \end{bmatrix} \begin{bmatrix} h_1 \\ h_2 \end{bmatrix} \quad (4.28)$$

The horizontal component of velocity of scallop B due to the f_x^A dipole of A is:

$$\begin{aligned} V_x^{BA_x} &= \frac{d}{8\pi\mu} h_1 [G_{11}^A \quad G_{12}^A] \begin{bmatrix} \dot{\alpha}_{1A} \\ \dot{\alpha}_{2A} \end{bmatrix} \\ &= [V_{11}^{BA_x} \quad V_{12}^{BA_x}] \begin{bmatrix} \dot{\alpha}_{1A} \\ \dot{\alpha}_{2A} \end{bmatrix} \end{aligned}$$

Similarly the vertical component of velocity is:

$$V_y^{BA} = [V_{21}^{BA_x} \quad V_{22}^{BA_x}] \begin{bmatrix} \dot{\alpha}_{1A} \\ \dot{\alpha}_{2A} \end{bmatrix}$$

Total velocity at B due to f_x^A force dipole:

$$\begin{bmatrix} V_x^{BA_x} \\ V_y^{BA_x} \end{bmatrix} = \begin{bmatrix} V_{11}^{BA_x} & V_{12}^{BA_x} \\ V_{21}^{BA_x} & V_{22}^{BA_x} \end{bmatrix} \begin{bmatrix} \dot{\alpha}_{1A} \\ \dot{\alpha}_{2A} \end{bmatrix} = V^{BA_x}(\alpha_{1A}, \alpha_{2A}, x_a, x_b, y_a, y_b) \begin{bmatrix} \dot{\alpha}_{1A} \\ \dot{\alpha}_{2A} \end{bmatrix} \quad (4.29)$$

Similarly, velocity at B due to f_y^A force dipole:

$$\begin{bmatrix} V_x^{BA_y} \\ V_y^{BA_y} \end{bmatrix} = \begin{bmatrix} V_{11}^{BA_y} & V_{12}^{BA_y} \\ V_{21}^{BA_y} & V_{22}^{BA_y} \end{bmatrix} \begin{bmatrix} \dot{\alpha}_{1A} \\ \dot{\alpha}_{2A} \end{bmatrix} \quad (4.30)$$

Velocity at B due to A 's hydrodynamic field is the superposition of 4.29 and 4.30:

$$\begin{bmatrix} V_x^{BA} \\ V_y^{BA} \end{bmatrix} = \begin{bmatrix} V_{11}^{BA} & V_{12}^{BA} \\ V_{21}^{BA} & V_{22}^{BA} \end{bmatrix} \begin{bmatrix} \dot{\alpha}_{1A} \\ \dot{\alpha}_{2A} \end{bmatrix} = V^{BA}(\alpha_{1A}, \alpha_{2A}, x_a, x_b, y_a, y_b) \begin{bmatrix} \dot{\alpha}_{1A} \\ \dot{\alpha}_{2A} \end{bmatrix} \quad (4.31)$$

Similarly velocity at A due to B's hydrodynamic field:

$$\begin{bmatrix} V_x^{AB} \\ V_y^{AB} \end{bmatrix} = V^{AB}(\alpha_{1B}, \alpha_{2B}, x_a, x_b, y_a, y_b) \begin{bmatrix} \dot{\alpha}_{1B} \\ \dot{\alpha}_{2B} \end{bmatrix} \quad (4.32)$$

However, for the purpose of ease of computation, the velocity at B due to A, is computed by overlapping the Stokeslets generated by individual links of A using 4.3. This yields similar results as utilising the formula for Stokes Dipole due to A, only advantage being reduced computational complexity. Similarly for computing the hydrodynamic field of B.

4.3 Two-microscallop coupled model

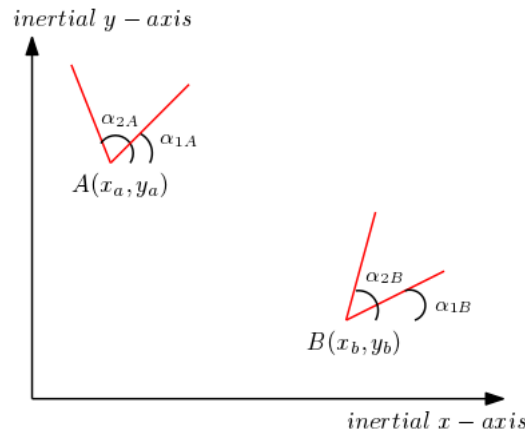


FIGURE 4.6: Two-microscallop coupled model

The final state of the above system is 8 dimensional, whose dynamics is given by 4.33. The expressions for each of these terms have been determined in the previous sections.

$$\begin{bmatrix} \dot{\alpha}_{1A} \\ \dot{\alpha}_{2A} \\ \dot{x}_A \\ \dot{y}_A \\ \dot{\alpha}_{1B} \\ \dot{\alpha}_{2B} \\ \dot{x}_B \\ \dot{y}_B \end{bmatrix} = \begin{bmatrix} \dot{\alpha}_{1A} \\ \dot{\alpha}_{2A} \\ K_{11}^A \dot{\alpha}_{1A} + K_{12}^A \dot{\alpha}_{2A} \\ K_{21}^A \dot{\alpha}_{1A} + K_{22}^A \dot{\alpha}_{2A} \\ 0 \\ 0 \\ V_{11}^{BA} \dot{\alpha}_{1A} + V_{12}^{BA} \dot{\alpha}_{2A} \\ V_{21}^A \dot{\alpha}_{1A} + V_{22}^A \dot{\alpha}_{2A} \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ V_{11}^{AB} \dot{\alpha}_{1B} + V_{12}^{AB} \dot{\alpha}_{2B} \\ V_{21}^{AB} \dot{\alpha}_{1B} + V_{22}^{AB} \dot{\alpha}_{2B} \\ \dot{\alpha}_{1B} \\ \dot{\alpha}_{2B} \\ K_{11}^B \dot{\alpha}_{1B} + K_{12}^B \dot{\alpha}_{2B} \\ K_{21}^B \dot{\alpha}_{1B} + K_{22}^B \dot{\alpha}_{2B} \end{bmatrix}$$

$$\begin{bmatrix} \dot{x}_A \\ \dot{y}_A \\ \dot{\alpha}_{1A} \\ \dot{\alpha}_{2A} \\ \dot{x}_B \\ \dot{y}_B \\ \dot{\alpha}_{1B} \\ \dot{\alpha}_{2B} \end{bmatrix} = \begin{bmatrix} K_{11}^A & K_{12}^A & V_{11}^{AB} & V_{12}^{AB} \\ K_{21}^A & K_{22}^A & V_{21}^{AB} & V_{22}^{AB} \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ V_{11}^{BA} & V_{12}^{BA} & K_{11}^B & K_{12}^B \\ V_{21}^{BA} & V_{22}^{BA} & K_{21}^B & K_{22}^B \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{\alpha}_{1A} \\ \dot{\alpha}_{2A} \\ \dot{\alpha}_{1B} \\ \dot{\alpha}_{2B} \end{bmatrix} \quad (4.33)$$

With the above model in place, it is necessary to validate it through simulations of certain test cases.

4.4 Model validation - simulations

This section presents the evolution of the above system when actuated as described in the following three scenarios.

1. When two scallops in close proximity undergo motion with no phase difference, it results in the hydrodynamic interactions also being symmetrical. Hence the system succumbs to the reciprocal theorem and the net displacement of the entire system is zero.

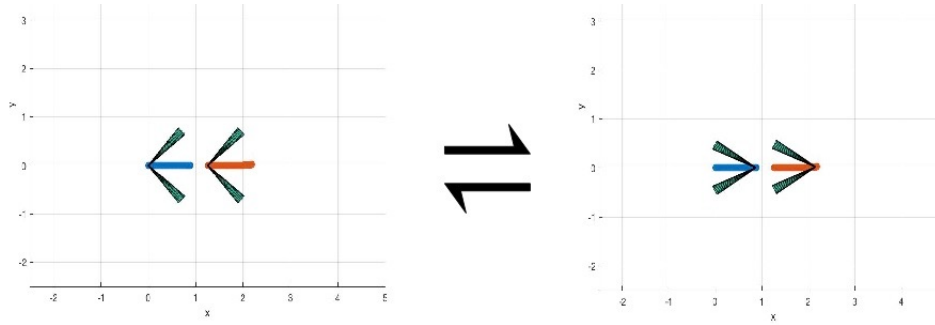
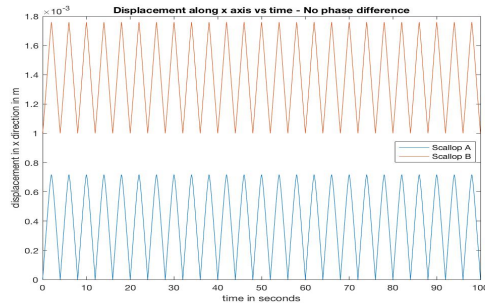
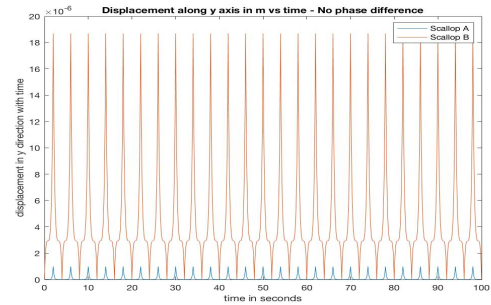


FIGURE 4.7: Two end configurations of the system

The variations in the x , y and the scallop links are as below:



Motion along x axis.



Motion along y axis.

FIGURE 4.8: Displacement in m against seconds

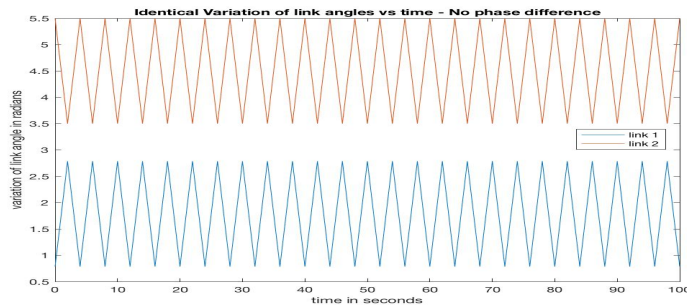


FIGURE 4.9: Variation of the link angles - identical for both scallops as the phase difference between them is zero.

2. When the scallops undergoes reciprocal motion, with non trivial phase differences, the reciprocity is broken, and the system does possess a non zero displacement.

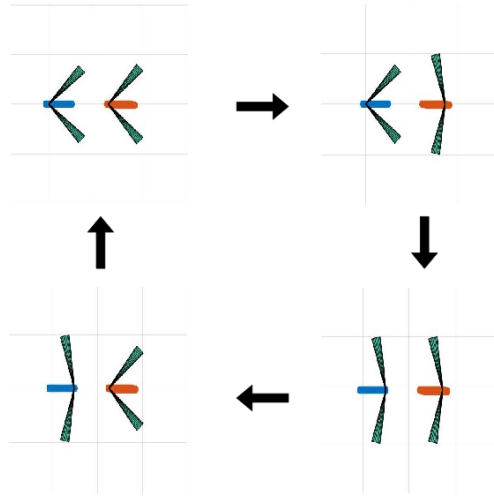
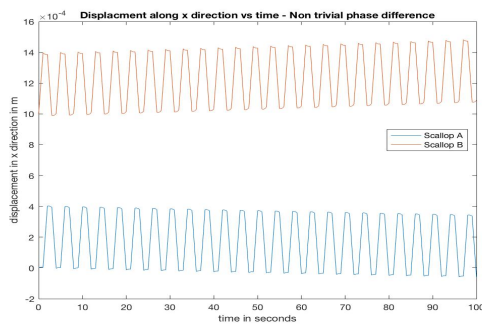
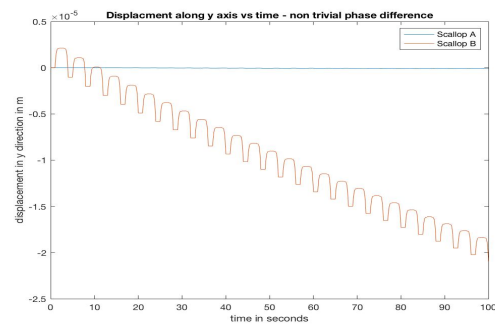


FIGURE 4.10: The cyclic change of configurations of the system

The variations in the x , y and the scallop links are as below:



Motion along x axis.



Motion along y axis.

FIGURE 4.11: Displacement in m against seconds

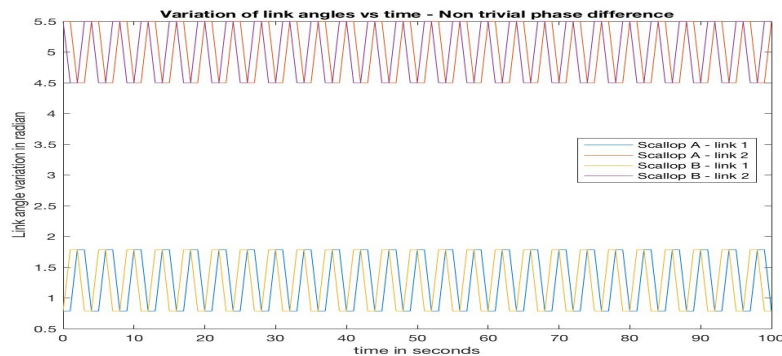


FIGURE 4.12: Variation of the link angles of both the scallops

The bias in the x and y displacements are evident, due to the breakdown of the scallop theorem.

3. When the scallops are considerably away from one another, there is negligible coupling between the two. In that case, when actuated with non trivial phase difference, the scallops still do not undergo any non zero displacement.

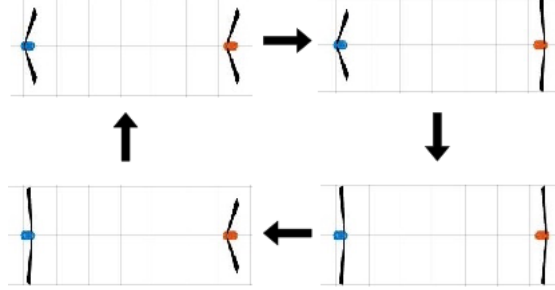


FIGURE 4.13: The cyclic change of configurations of the system

The variations in the x , y and the scallop links are as below:

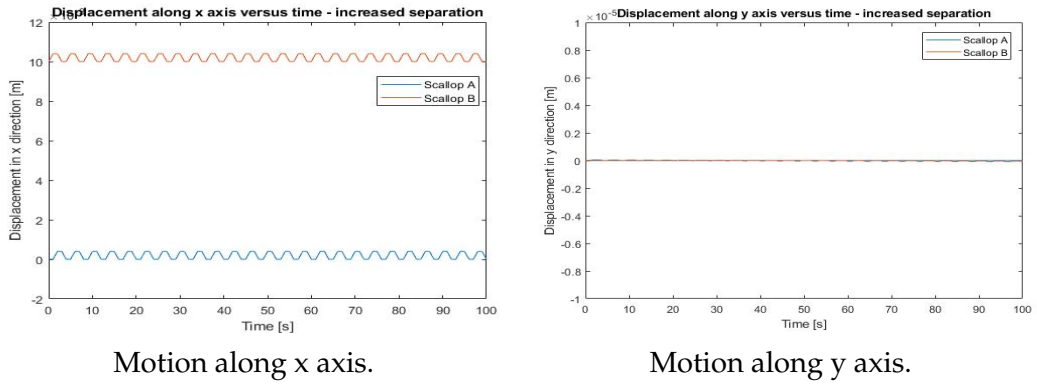


FIGURE 4.14: Displacement in m against seconds

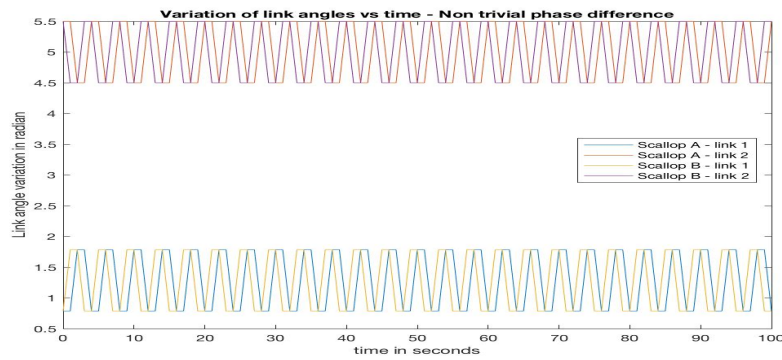


FIGURE 4.15: Variation of the link angles of both the scallops.

Post model validation using test cases, it is necessary to be able to control and steer this system to our desired configurations. With this intention, the next chapter focuses on generating control laws and motion planning of the kinematic system to achieve the desired states.

Chapter 5

Motion planning and control

This chapter mainly presents the results regarding the controllability of 4.33. We then move on to devise a control law via a Lyapunov function technique to achieve a certain objective. The chapter then concludes with a detailed description of the differential geometric motion planning implementation proposed in Lafferriere and Sussmann, 1993 to steer the system from an initial to a desired configuration.

5.1 Controllability

Before we attempt to synthesize control laws or implement motion planning algorithms, it is necessary to check the controllability of the system. With the definition of reachable sets defined for general systems in 3.6, we move onto define the notion of *small time locally controllability* (STLC).

Definition 5.1.1 Let $\Sigma = (\mathcal{M}, \mathcal{C} = \{f_1, \dots, f_m\}, \mathcal{U})$ be a C^∞ control affine system and let $x_0 \in \mathcal{M}$, then Σ is small time locally controllable (STLC) from x_0 if there exists a $T > 0$ such that $x_0 \in \text{int}(\mathcal{R}_\Sigma(x_0, \leq t))$ for each $t \in [0, T]$.

The pair $(\mathcal{M}, \mathcal{C})$ is properly small time locally controllable (properly STLC) if $(\mathcal{M}, \mathcal{C}, \mathcal{U})$ is STLC for every proper control set $\mathcal{U}$.

The controllability of a driftless control-affine system can be characterized using the Lie algebra rank condition (or LARC)(Bullo and Lewis, 2004).

Theorem 5.1.1 Let $\mathcal{C} = \{f_0, f_1, \dots, f_m\}$ be C^∞ vector fields on $\mathcal{M}$ with f_0 identically zero. The pair $(\mathcal{M}, \mathcal{C})$ is properly STLC from x_0 if $\text{Lie}^{(\infty)}(\mathcal{C})_{x_0} = T_{x_0}\mathcal{M}$.

The system in 4.33 can be rewritten as:

$$\dot{x}(t) = f_1(x(t))u_1 + f_2(x(t))u_2 + f_3(x(t))u_3 + f_4(x(t))u_4$$

where $x(t)$ is a curve in an eight dimensional state space $\mathcal{M}$ (corresponding to the two sets of states of the scallop A and B), u_1, u_2, u_3 and u_4 are the control inputs which are the angular velocities of the links, that is, $\alpha_{1A}, \alpha_{2A}, \alpha_{1B}$ and α_{2B} respectively. The control inputs $\{u_1, \dots, u_4\} \in U$, where U is a subset in $\mathbb{R}^4$. $f_1, \dots, f_4$ are the control vector fields through which the control input influences the state of the system.

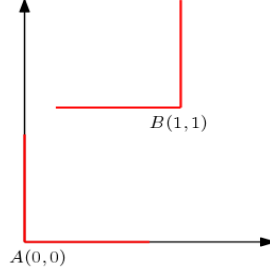


FIGURE 5.1: Configuration: $x_A = 0, y_A = 0, \alpha_{1A} = 0, \alpha_{2A} = 1.57,$
 $x_B = 1, y_B = 1, \alpha_{1B} = 1.57$ and $\alpha_{2B} = 3.14$

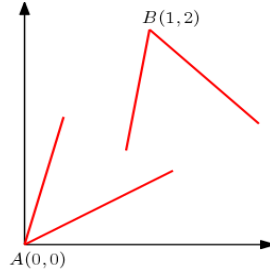


FIGURE 5.2: Configuration: $x_A = 0, y_A = 0, \alpha_{1A} = 0.79, \alpha_{2A} = 1.32,$
 $x_B = 1, y_B = 2, \alpha_{1B} = 5.5$ and $\alpha_{2B} = 4.62$

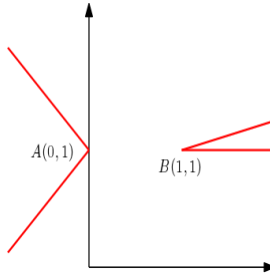


FIGURE 5.3: Configuration: $x_A = 0, y_A = 1, \alpha_{1A} = 2.36, \alpha_{2A} = 3.93,$
 $x_B = 1, y_B = 1, \alpha_{1B} = 0$ and $\alpha_{2B} = 0.52$

The Lie brackets computed for the purpose of small time local controllability evaluation were

$$[f_1, f_2], [f_2, f_3], [f_3, f_4] \text{ \& } [f_1, f_3]$$

These brackets along with f_1, f_2, f_3 and f_4 were found to span $T_{x_0}\mathcal{M}$. All the above symbolic computations were conducted on Mathematica. For values of these brackets for one of these configurations, please refer to [B](#). Hence, it can be concluded from above that Σ is small time locally controllable from these locations and possible other configurations as well.

Now that we have established the coordinated scallop system is STLCL from most of the configurations, we proceed to devise a control law using Lyapunov functions in order to achieve a certain objective.

5.2 Lyapunov function - control law generation

One of the common objectives in a multibot system is to bring them to an identical state from various initial conditions. Often, this translates to physically bringing them together to a common position in space from various respective positions. This section delineates the generation of a control law via the Lyapunov function approach in order to drive the two scallops close to one another. A similar approach can be employed to reach other configurations.

Let the Scallops A and B be located at (x_a, y_a) and (x_b, y_b) respectively. Then the candidate Lyapunov function V intended to drive them is chosen as:

$$V = (x_a - x_b)^2 + (y_a - y_b)^2 \quad (5.1)$$

The control law adopted should be such that the system approaches an asymptotically stable point. The derivative of the Lyapunov function is as follows:

$$\dot{V} = 2[(x_a - x_b)(\dot{x}_a - \dot{x}_b) + (y_a - y_b)(\dot{y}_a - \dot{y}_b)]$$

Substituting expressions in 4.33 for the velocity terms,

$$\begin{aligned} \dot{V} = & 2[\alpha_{1A}\{(x_a - x_b)(K_{11}^A - V_{11}^{BA}) + (y_a - y_b)(K_{21}^A - V_{21}^{BA})\} \\ & + \alpha_{2A}\{(x_a - x_b)(K_{12}^A - V_{12}^{BA}) + (y_a - y_b)(K_{22}^A - V_{22}^{BA})\} \\ & + \alpha_{1B}\{(x_a - x_b)(V_{11}^{AB} - K_{11}^B) + (y_a - y_b)(V_{21}^{AB} - K_{21}^B)\} \\ & + \alpha_{2B}\{(x_a - x_b)(V_{12}^{AB} - K_{12}^B) + (y_a - y_b)(V_{22}^{AB} - K_{22}^B)\}] \end{aligned} \quad (5.2)$$

In order to obtain $\dot{V} \leq 0$, we need the control inputs to be in the following form:

$$\begin{aligned} \alpha_{1A} &= -c_1\{(x_a - x_b)(K_{11}^A - V_{11}^{BA}) + (y_a - y_b)(K_{21}^A - V_{21}^{BA})\} \\ \alpha_{2A} &= -c_2\{(x_a - x_b)(K_{12}^A - V_{12}^{BA}) + (y_a - y_b)(K_{22}^A - V_{22}^{BA})\} \\ \alpha_{1B} &= -c_3\{(x_a - x_b)(V_{11}^{AB} - K_{11}^B) + (y_a - y_b)(V_{21}^{AB} - K_{21}^B)\} \\ \alpha_{2B} &= -c_4\{(x_a - x_b)(V_{12}^{AB} - K_{12}^B) + (y_a - y_b)(V_{22}^{AB} - K_{22}^B)\} \end{aligned}$$

where c_1, c_2, c_3 and c_4 are positive.

With the above expressions for control input, final expression for $\dot{V}$ is:

$$\dot{V} = -2\{c_1\alpha_{1A}^2 + c_2\alpha_{2A}^2 + c_3\alpha_{1B}^2 + c_4\alpha_{2B}^2\} \leq 0 \quad (5.3)$$

We now adopt LaSalle's theorem (stated below) to draw a conclusion (Khalil, 2002).

Theorem 5.2.1 For the system $\dot{x} = f(x)$, let $\Omega \in D$ be a compact set that is positively invariant. Let $V : D \rightarrow R$ be a continuously differentiable function such that $\dot{V}(x) \leq 0$ in Ω . Let E be the set of all points in Ω where $\dot{V}(x) = 0$. Let M be the largest invariant set in E . The every solution starting in Ω approaches M as $t \rightarrow \infty$.

The set E in our case is defined as:

$$E = \{x \in D | \dot{V}(x) = 0\}$$

From 5.3, $\dot{V} \equiv 0$ only when all the control inputs are 0. In order for this to occur, either the positions of the scallop hinges need to converge, that is, $x_a = x_b$ and $y_a = y_b$ or $K_{ij}^k = V_{ij}^{lk}$. The latter involve a complicated nonlinear expression identically going to zero, which we rule out more from an intuitive examination. Hence, under

the influence of the above control input, the scallops asymptotically approach each other (the grouping up of the scallops.) The phenomenon described here can be observed in the following simulation results:

- Initial position: $x_A = 0, y_A = 0, \alpha_{1A} = 0.79, \alpha_{2A} = 5.5, x_B = 0.001, y_B = 0.0005, \alpha_{1B} = 1.57$ and $\alpha_{2B} = 5.5$. ($c_1 = c_2 = c_3 = c_4 = 10^5$)

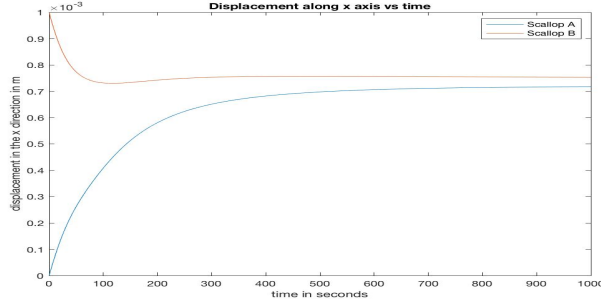


FIGURE 5.4: X-displacement with time

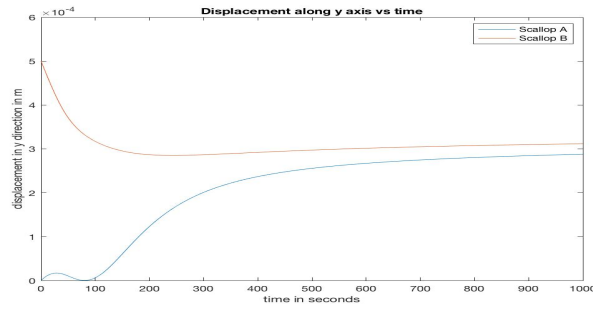


FIGURE 5.5: Y-displacement with time

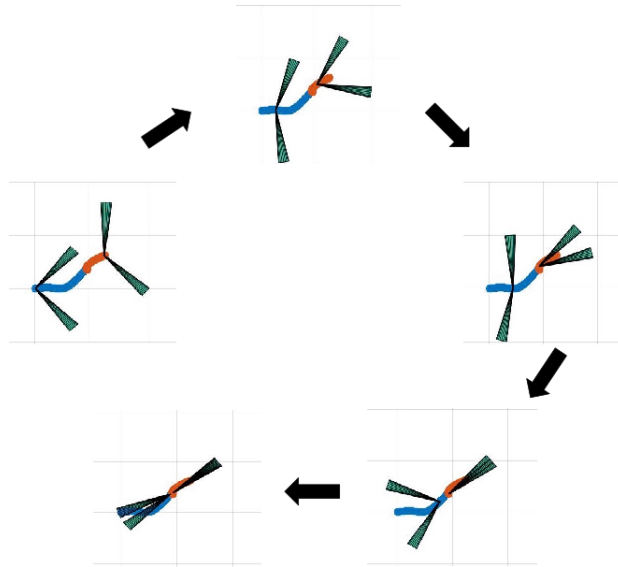


FIGURE 5.6: Evolution of scallop configurations

- Initial position: $x_A = 0.0005$, $y_A = 0.0002$, $\alpha_{1A} = 2.1$, $\alpha_{2A} = 3.93$, $x_B = -0.0005$, $y_B = -0.0005$, $\alpha_{1B} = 0.6$ and $\alpha_{2B} = 5.5$. ($c_1 = c_2 = c_3 = c_4 = 10^5$)

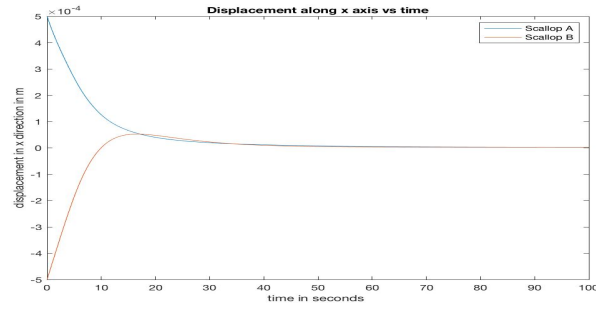


FIGURE 5.7: X-displacement with time

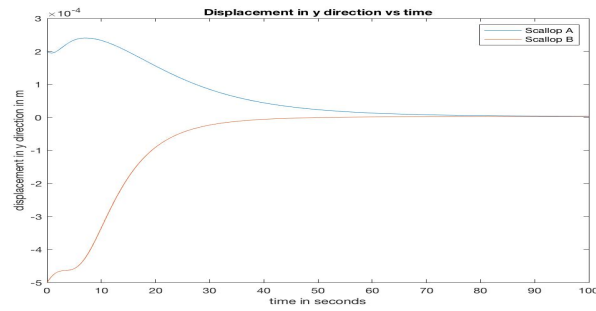


FIGURE 5.8: Y-displacement with time

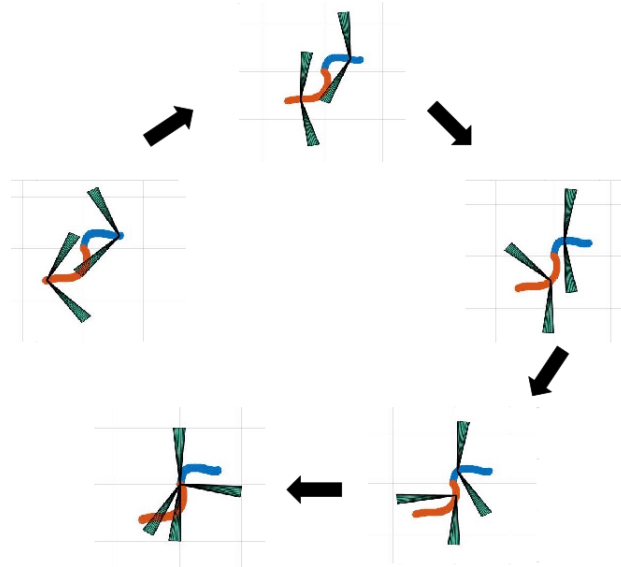


FIGURE 5.9: Evolution of scallop configurations

5.3 Motion planning

Motion planning for nonlinear systems is a non-trivial task. Very few techniques exist for even driftless systems. Since the system in 4.33 is driftless, we utilise the differential geometric approach for motion planning provided by Lafferriere and Sussmann, 1993. The main advantage of this algorithm is that it doesn't impose excessive restrictions on the system, apart from an evolution on a Lie group. It provides exact convergence to the desired configuration, provided the system is nilpotent, and if it isn't, it converges approximately close to what is desired. Under certain conditions, the final error in convergence is bounded as well.

The intention of this algorithm is to steer the system from one configuration state to another in a fixed amount of time. Since the system is small time locally controllable as proved in 5.1, there are control inputs available to reach any configuration given an initial state. This algorithm provides with one such set of controls, which although not as computationally intensive as others to deduce, involve a lot of moves as will be showcased in the upcoming implementation. But due to its lack of dependency on generators, this algorithm can be used with a fair degree of accuracy to help achieve the final state for almost all driftless systems.

The preliminaries required to understand this algorithm and the description of the same is illustrated in detail, along with an example in A. We present the direct application of the algorithm to our system further ahead.

Consider the system given in 4.33, representing the coupled scallop kinematics. The system is not nilpotent. There are four control inputs and therefore 4 control vector fields, namely, f_1, f_2, f_3 and f_4 . These control fields will correspond to X_1, X_2, X_3 and X_4 in the polynomial space respectively. Further analysis that will be conducted will be along the lines of a system whose order of nilpotency is 2, as the number of basis vectors such a Lie algebra are sufficient to span the tangent space of the configuration manifold.

The first step is to construct a curve such that $\gamma(0) = p$ and $\gamma(T) = q$.

Initial configuration (p): $(x_a^i, y_a^i, \alpha_{1A}^i, \alpha_{2A}^i, x_b^i, y_b^i, \alpha_{1B}^i, \alpha_{2B}^i)$

Final configuration (q): $(x_a^f, y_a^f, \alpha_{1A}^f, \alpha_{2A}^f, x_b^f, y_b^f, \alpha_{1B}^f, \alpha_{2B}^f)$

$$\gamma(t) = \begin{bmatrix} x_a^i + t \left(\frac{x_a^f - x_a^i}{T} \right) \\ y_a^i + t \left(\frac{y_a^f - y_a^i}{T} \right) \\ \alpha_{1A}^i + t \left(\frac{\alpha_{1A}^f - \alpha_{1A}^i}{T} \right) \\ \alpha_{2A}^i + t \left(\frac{\alpha_{2A}^f - \alpha_{2A}^i}{T} \right) \\ x_b^i + t \left(\frac{x_b^f - x_b^i}{T} \right) \\ y_b^i + t \left(\frac{y_b^f - y_b^i}{T} \right) \\ \alpha_{1B}^i + t \left(\frac{\alpha_{1B}^f - \alpha_{1B}^i}{T} \right) \\ \alpha_{2B}^i + t \left(\frac{\alpha_{2B}^f - \alpha_{2B}^i}{T} \right) \end{bmatrix} = \begin{bmatrix} x_a(t) \\ y_a(t) \\ \alpha_{1A}(t) \\ \alpha_{2A}(t) \\ x_b(t) \\ y_b(t) \\ \alpha_{1B}(t) \\ \alpha_{2B}(t) \end{bmatrix} \quad (5.4)$$

The derivative of the curve is given by:

$$\gamma(t) = \frac{1}{T} \begin{bmatrix} x_a^f - x_a^i \\ y_a^f - y_a^i \\ \alpha_{1A}^f - \alpha_{1A}^i \\ \alpha_{2A}^f - \alpha_{2A}^i \\ x_b^f - x_b^i \\ y_b^f - y_b^i \\ \alpha_{1B}^f - \alpha_{1B}^i \\ \alpha_{2B}^f - \alpha_{2B}^i \end{bmatrix}$$

The extended system is given by:

$$\dot{\gamma}(t) = \sum_{j=1}^{10} f_j v_j \quad (5.5)$$

The corresponding system in the polynomial space is:

$$\dot{S}_e(t) = S_e \sum_{j=1}^{10} B_j v_j \quad (5.6)$$

where,

$$B_1 \equiv X_1, \quad B_2 \equiv X_2, \quad B_3 \equiv X_3, \quad B_4 \equiv X_4, \quad B_5 \equiv [X_1, X_2],$$

$$B_6 \equiv [X_2, X_3], \quad B_7 \equiv [X_3, X_4], \quad B_8 \equiv [X_1, X_3], \quad B_9 \equiv [X_1, X_4], \quad \& \quad B_{10} \equiv [X_2, X_4]$$

The states are expressed in terms of the coordinates of the curve in 5.4. Since we need only 8 linearly independent vector fields to span the vector space of the manifold, in 5.5 v_9 and v_{10} are assumed to be zero (the vector fields f_1 to f_8 are linearly independent). Hence we obtain a square matrix in the place of the original rectangular one. By inverting the matrix in 5.5, we get the values for v . All the above calculations are performed on Mathematica. The solution to the system in 5.6 takes on the form:

$$S(t) = e^{h_{10}B_{10}} e^{h_9B_9} e^{h_8B_8} e^{h_7B_7} e^{h_6B_6} e^{h_5B_5} e^{h_4B_4} e^{h_3B_3} e^{h_1B_1}$$

Taking the derivative of the solution yields,

$$\begin{aligned} \dot{S}(t) = S(t) & (\dot{h}_1 B_1 + e^{-h_1 B_1} B_2 e^{h_1 B_1} \dot{h}_2 + e^{-h_1 B_1} e^{-h_2 B_2} B_3 e^{h_2 B_2} e^{h_1 B_1} \dot{h}_3 + e^{-h_1 B_1} e^{-h_2 B_2} \\ & e^{-h_3 B_3} B_4 e^{h_3 B_3} e^{h_2 B_2} e^{h_1 B_1} \dot{h}_4 + e^{-h_1 B_1} e^{-h_2 B_2} e^{-h_3 B_3} e^{-h_4 B_4} B_5 e^{h_4 B_4} e^{h_3 B_3} e^{h_2 B_2} e^{h_1 B_1} \dot{h}_5 \\ & + e^{-h_1 B_1} e^{-h_2 B_2} e^{-h_3 B_3} e^{-h_4 B_4} e^{-h_5 B_5} B_6 e^{h_5 B_5} e^{h_4 B_4} e^{h_3 B_3} e^{h_2 B_2} e^{h_1 B_1} \dot{h}_6 + e^{-h_1 B_1} e^{-h_2 B_2} e^{-h_3 B_3} e^{-h_4 B_4} \\ & e^{-h_5 B_5} e^{-h_6 B_6} B_7 e^{h_6 B_6} e^{h_5 B_5} e^{h_4 B_4} e^{h_3 B_3} e^{h_2 B_2} e^{h_1 B_1} \dot{h}_7 + e^{-h_1 B_1} e^{-h_2 B_2} e^{-h_3 B_3} e^{-h_4 B_4} \\ & e^{-h_5 B_5} e^{-h_6 B_6} e^{-h_7 B_7} B_8 e^{h_7 B_7} e^{h_6 B_6} e^{h_5 B_5} e^{h_4 B_4} e^{h_3 B_3} e^{h_2 B_2} e^{h_1 B_1} \dot{h}_8 + e^{-h_1 B_1} \\ & e^{-h_2 B_2} e^{-h_3 B_3} e^{-h_4 B_4} e^{-h_5 B_5} e^{-h_6 B_6} e^{-h_7 B_7} e^{-h_8 B_8} B_9 e^{h_8 B_8} e^{h_7 B_7} e^{h_6 B_6} e^{h_5 B_5} e^{h_4 B_4} e^{h_3 B_3} e^{h_2 B_2} e^{h_1 B_1} \dot{h}_9 \\ & + e^{-h_1 B_1} e^{-h_2 B_2} e^{-h_3 B_3} e^{-h_4 B_4} e^{-h_5 B_5} e^{-h_6 B_6} e^{-h_7 B_7} e^{-h_8 B_8} e^{-h_9 B_9} B_{10} e^{h_9 B_9} e^{h_7 B_7} e^{h_6 B_6} e^{h_5 B_5} e^{h_4 B_4} \\ & e^{h_3 B_3} e^{h_2 B_2} e^{h_1 B_1} \dot{h}_{10} \end{aligned} \quad (5.7)$$

An important property of the exponential mapping which is used to simplify the above expression is:

$$e^{-X}Ye^X = \sum_{n=0}^{\infty} (-1)^n \frac{(adX)^n}{n!} Y$$

Upon equating 5.7 to 5.6, we get the following set of O.D.E.s in the P. Hall coordinates corresponding to ν obtained from 5.5:

$$\begin{aligned} \dot{h}_1 &= \nu_1 \\ \dot{h}_2 &= \nu_2 \\ \dot{h}_3 &= \nu_3 \\ \dot{h}_4 &= \nu_4 \\ \dot{h}_5 &= \nu_5 + h_1 \dot{h}_2 = \nu_5 + h_1 \nu_2 \\ \dot{h}_6 &= \nu_6 + h_2 \dot{h}_3 = \nu_6 + h_2 \nu_3 \\ \dot{h}_7 &= \nu_7 + h_3 \dot{h}_4 = \nu_7 + h_3 \nu_4 \\ \dot{h}_8 &= \nu_8 + h_1 \dot{h}_3 = \nu_8 + h_1 \nu_3 \\ \dot{h}_9 &= h_1 \dot{h}_4 = h_1 \nu_4 \\ \dot{h}_{10} &= h_2 \dot{h}_4 = h_2 \nu_4 \end{aligned} \tag{5.8}$$

Solving 5.8 will yield us the expression for the P. Hall coordinates. Higher brackets of $S_e = S$ are expressed in terms of the original vector fields through the Campbell-Baker-Hausdorff formula as:

$$e^{a[X,Y]} = \begin{cases} e^{\sqrt{a}X} e^{\sqrt{a}Y} e^{-\sqrt{a}X} e^{-\sqrt{a}Y} & \text{if } a > 0 \\ e^{\sqrt{a}Y} e^{\sqrt{a}X} e^{-\sqrt{a}Y} e^{-\sqrt{a}X} & \text{if } a < 0 \end{cases} \tag{5.9}$$

The values of these P. Hall coordinates at time T , along with 5.9 will provide us with the necessary sequence of control inputs. Consider the following example:

Initial configuration: $x_A = 0m, y_A = 0m, \alpha_{1A} = 0.35rad, \alpha_{2A} = 1.57rad, x_B = 0.002m, y_B = 0.002m, \alpha_{1B} = 0.79rad$ and $\alpha_{2B} = 2.36rad$

Final configuration: $x_A = 0.0001m, y_A = -0.0001m, \alpha_{1A} = 1.05rad, \alpha_{2A} = 2.62rad, x_B = 0.0025m, y_B = 0.0025m, \alpha_{1B} = 1.57rad$ and $\alpha_{2B} = 3.31rad$

Time duration: 100s

The P. Hall coordinates for the required steering is: $h_1 = 0.7, h_2 = 1.05, h_3 = 0.79, h_4 = 0.96, h_5 = 0.59, h_6 = -43.74, h_7 = -3.65, h_8 = 956.34, h_9 = 0.34, h_{10} = 0.5$

With the above coordinates, the final solution to the differential equation 5.6 is:

$$\begin{aligned} S(t) = & e^{0.7X_2} e^{0.7X_4} e^{-0.7X_2} e^{-0.7X_4} e^{0.58X_1} e^{0.58X_4} e^{-0.58X_1} e^{-0.58X_4} e^{30.93X_1} e^{30.93X_3} \\ & e^{-30.93X_1} e^{-30.93X_3} e^{1.91X_4} e^{1.91X_3} e^{-1.91X_4} e^{-1.91X_3} e^{6.61X_3} e^{6.61X_2} e^{-6.61X_3} \\ & e^{-6.61X_2} e^{0.77X_1} e^{0.77X_2} e^{-0.77X_1} e^{-0.77X_2} e^{0.96X_4} e^{0.79X_3} e^{1.05X_2} e^{0.7X_1} \end{aligned} \tag{5.10}$$

The sequence of control input which corresponds to the flow given in 5.10 is tabularised as:

u_1	u_2	u_3	u_4	t
0	1	0	0	0.71
0	0	0	1	1.42
0	-1	0	0	2.13
0	0	0	-1	2.84
1	0	0	0	3.42
0	0	0	1	4
-1	0	0	0	4.58
0	0	0	-1	5.16
1	0	0	0	36.09
0	0	1	0	67.02
-1	0	0	0	97.95
0	0	-1	0	128.88
0	0	0	1	130.79
0	0	1	0	132.7
0	0	0	-1	134.61
0	0	-1	0	136.52
0	0	1	0	143.13
0	1	0	0	149.74
0	0	-1	0	156.35
0	-1	0	0	162.96
1	0	0	0	163.73
0	1	0	0	164.5
-1	0	0	0	165.27
0	-1	0	0	166.04
0	0	0	1	167
0	0	1	0	167.79
0	1	0	0	168.84
1	0	0	0	169.54

TABLE 5.1: Sequence of input values

It can be seen above that the total time is 169.54s. In order to scale it to 100s, we multiply the time duration with $\frac{100}{169.54}$ and scale the control values to $\frac{169.54}{100}$ times their magnitude. The final figures corresponding to the evolution under such control input values are as below.

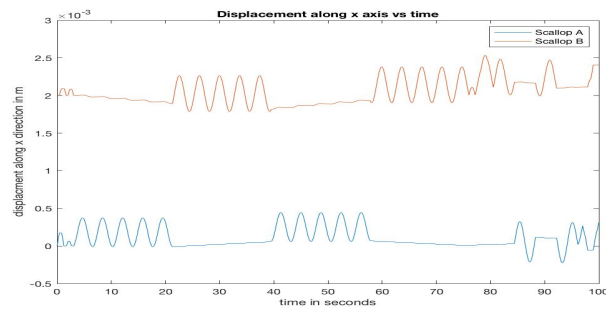


FIGURE 5.10: X-displacement with time

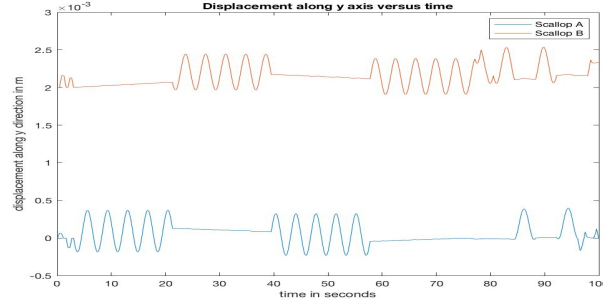


FIGURE 5.11: Y-displacement with time

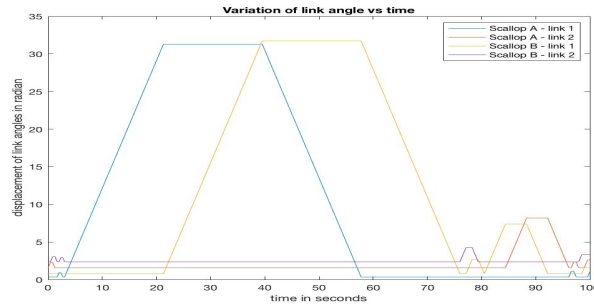


FIGURE 5.12: Link angles with time

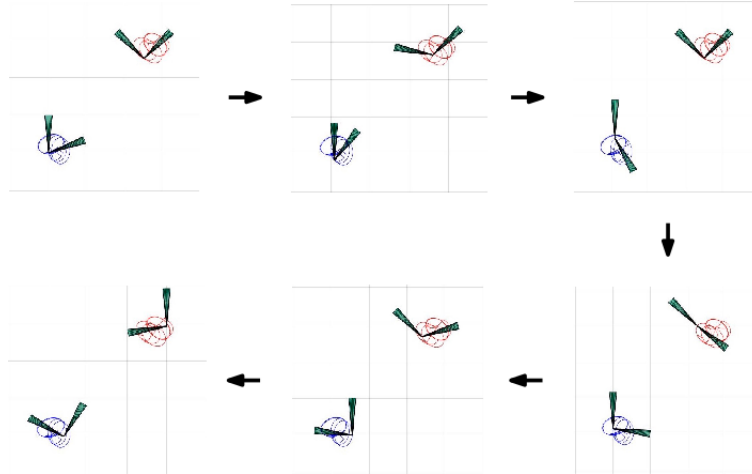


FIGURE 5.13: Evolution of scallop configurations

Post simulations, the final values of the state space were as follows: $x_A = 0.0003m$, $y_A = -0.000003m$, $\alpha_{1A} = 1.05rad$, $\alpha_{2A} = 2.62rad$, $x_B = 0.0024m$, $y_B = 0.0023m$, $\alpha_{1B} = 1.57rad$ and $\alpha_{2B} = 3.31rad$. The angles of the links converge to the exact value as desired, however the actual positions are fairly close to the expected values but not exact, due to the lack of nilpotency. With smaller displacement per every maneuver, it is possible to attain every configuration in the state space. We are able to steer to our desired location with a fair degree of accuracy and hence this motion planning technique can be utilised for realising desired configurations.

This chapter concludes with the above simulation and results pertaining to motion planning of the system. We next discuss the possible future work in this arena of microrobotics.

Chapter 6

Conclusion and future work

This thesis explores locomotion-related features of mechanisms moving in the LRN regime. It primarily focuses on overcoming the scallop theorem, which states that reciprocal motion in LRN in Newtonian fluids, leads to no net displacements. However, reciprocal motion with asymmetrical rates of strokes in NN fluids leads to asymmetrical forces acting on the scallop leading to a non zero displacement. Hence, this particular aspect is exploited in the study of behaviour of microscallop in NN fluids. In conjunction to the above, another mechanism which can be exploited to overcome the scallop theorem is the one of coordinated motion. Reciprocal motion of individual scallops with non-zero phase differences are subject to the hydrodynamic forces of one another. This leads to a break in symmetry with respect the displacements undergone in opposite directions. Hydrodynamic influences are modelled using the theory of Stokeslets and Stokes dipoles, leading to a kinematic description of a two microscallop system. This model is verified for expected behaviour with respect to certain test cases. Further the controllability and certain control laws have been determined in order to achieve desired configurations. Another important aspect of microrobots presented in the thesis is motion planning of the two scallop system. A differential geometric approach proposed by Lafferriere and Sussmann, 1993, which is a generic method of motion planning for control-affine driftless systems - is employed to generate the sequence of control inputs necessary to steer the coordinated system from one position to another. The work in this thesis can be further extended by exploring other avenues of locomotion in microrobotics. Certain avenues of future work are:

1. Extending the two microscallop system to n microscallops and implementing motion planning techniques which account for collision avoidance (Kim, Julius, and Kim, 2017).
2. Study of such coordinated systems in NN fluids (Kim et al., 2017).
3. One of the most important concerns regarding such work is the computational complexity these models entail. This, by default imposes constraints on the degree to which systems can be modelled, and thereby controlled. Any improvements towards reducing the complexity without compromising accuracy will prove to be helpful in analyses and design by a long mile.
4. Apart from Newtonian and time independent NN fluids, study of such mechanisms in complex fluids can be of great utility. When we transcend into nanorobotics, blood, along with it's corpuscles should be modelled as a complex fluid. Although there is not much theory to characterise these fluids, any empirical fits to existing data can be performed to design and control locomotion techniques of microrobots in such environment (Campo-DeaÃo, 2016).

5. The swimmer links can be modelled as deformable rather than rigid, which can help in breaking the reciprocity. Other possible mechanisms (rotating filaments, etc) can be studied to design a swimmer that undergoes finite displacement under reciprocal actuation (Nelson, Sun, and Greminger, 2005).

Considering the many significant applications that microrobotics possesses in the biomedical field, especially in drug delivery, a study into these mechanisms will surely be of aid in the design of smart carriers. A strong theoretical ground along with the experiments will help in the robust control and better manipulation of such microsystems.

Appendix A

Differential geometric approach to motion planning

A.1 Preliminaries

Consider the following driftless system:

$$\dot{\vec{x}} = \sum_{j=1}^m g_j(x) u_j \quad (\text{A.1})$$

Definition A.1.1 *The above system is said to be nilpotent of order k if all the Lie brackets containing $k + 1$ vector fields vanish.*

In A.1, if $m = 2$, and if g_1, g_2 and $[g_1, g_2]$ are linearly independent at the operating point, x_0 , then $\exists$ a transformation $\vec{u} = [\beta(\vec{x})]\vec{w}$ such that the new system: $\dot{\vec{x}} = \tilde{g}_1 w_1 + \tilde{g}_2 w_2$ is nilpotent (Hermes, Lundell, and Sullivan, 1984). This is known as nilpotentisation.

Consider the Lie algebra of these vector fields denoted by $\text{Lie}^\infty(g_1, \dots, g_m)$. Since the system is nilpotent of order k , higher order brackets disappear and the Lie algebra can be represented as $\text{Lie}^k(g_1, \dots, g_m)$. Since the Lie algebra is a finite dimensional vector space with the Lie bracket, it is possible to define a basis. Due to the following properties of the Lie Bracket, the basis can be determined through a set of rules.

1. Anticommutativity:

$$[g_1, g_2] = -[g_2, g_1]$$

2. Jacobi Lie Identity

$$[g_1, [g_2, g_3]] + [g_3, [g_1, g_2]] + [g_2, [g_3, g_1]] = 0$$

Definition A.1.2 *Basis of the Lie algebra $\text{Lie}^k(g_1, \dots, g_m)$ is known as the P. Hall Basis which is an ordered set of Lie products $\mathcal{B}(m, k) = \{\mathcal{B}_i\}$ satisfying:*

- $g_i \in \mathcal{B}(m, k), i = 1, \dots, m$
- If $l(\mathcal{B}_i) < l(\mathcal{B}_j)$ then $\mathcal{B}_i < \mathcal{B}_j$
- $[\mathcal{B}_i, \mathcal{B}_j] \in \mathcal{B}(m, k)$ if and only if
 1. $\mathcal{B}_i, \mathcal{B}_j \in \mathcal{B}(m, k)$ and $\mathcal{B}_i < \mathcal{B}_j$
 2. either $\mathcal{B}_i = g_p$ for some p or $\mathcal{B}_j = [\mathcal{B}_l, \mathcal{B}_r]$ with $\mathcal{B}_l, \mathcal{B}_r \in \mathcal{B}(m, k)$ and $\mathcal{B}_l \leq \mathcal{B}_r$

Example A.1.1 When $m = 2$ and $k = 3$, the P. Hall basis is given by $\mathcal{B}(2, 3) = \{X, Y, [X, Y], [X, [X, Y]], [Y, [X, Y]]\}$

We next introduce the notion of an extended system. A.1 is assumed to be controllable (satisfying the Lie algebra rank condition). Then at any x_0 ,

$$\mathcal{S}(g_i) = \text{span}\{g_1, \dots, g_m, [g_i, g_j], \dots, \text{ad}_{g_i}^{(k-1)} g_j\}_{x_0} = \mathcal{R}^n$$

where

$$\mathcal{B}_1 = g_1, \dots, \mathcal{B}_m = g_m, \mathcal{B}_{m+1} = [g_i, g_j], \dots, \mathcal{B}_r = \text{ad}_{g_i}^{(k-1)} g_j$$

and $r \geq n$. Since we are considering these vectors at x_0 , some of them might vanish at other points, and therefore $\mathcal{S}(g_i) \subset \mathcal{B}(m, k) = \{\mathcal{B}_1, \dots, \mathcal{B}_m, \mathcal{B}_{m+1}, \dots, \mathcal{B}_r\}$. Then the extended system is defined as:

$$\dot{\mathcal{X}}_e = \sum_{j=1}^r \mathcal{B}_j(\mathcal{X}_e) v_j \quad (\text{A.2})$$

where

$$v_j = \begin{cases} u_j, & \text{if } j = 1, \dots, m \\ v_{m+1}, \dots, v_r, & \text{virtual inputs} \end{cases}$$

In order to implement the second step of the algorithm, it is necessary to invoke results from the theory of non-commutative polynomials (related to the theory of free Lie algebra).

Let $A(X_1, \dots, X_m)$ be the set of non commutative polynomials, where X_i are the variables. This implies

$$X \cdot Y \neq Y \cdot X$$

Since polynomials is a vector space, and by the property of their anticommutativity, we can define a Lie Bracket,

$$[X, Y] = X \cdot Y - Y \cdot X$$

It can be checked that this definition of Lie bracket satisfies all the required properties, that is, linearity, anticommutativity and Jacobi-Lie identity.

Since $A(X_1, \dots, X_m)$ is a vector space with a Lie Bracket, this provides us with a Lie algebra denoted by $L(X_1, \dots, X_m)$. Here, although we are focusing on polynomials, X_i are indeterminates. They can be matrices, vector fields or polynomials. In this Lie algebra, we can also introduce nilpotency, represented by $L_k(X_1, \dots, X_m)$. Any polynomial of degree $k + 1$ in L vanishes. For example

$$X_1^{k-1} X_2 X_4 = 0$$

A P. Hall basis $\mathcal{B}(m, k)$, can be introduced for this system, which depends on the number of indeterminates and the order of nilpotency.

We now move onto to define the *exponential map*.

Definition A.1.3 For every $Z \in L(X_1, \dots, X_m)$, the exponential map is defined as:

$$e^Z = 1 + Z + \frac{Z^2}{2!} + \dots$$

However this exponential map gets truncated if Z belongs to a nilpotent Lie algebra. All the powers of the polynomial greater than k vanish and we are left with finite terms in the series.

Since we have an exponential map, we can now define a differential equation

$$\dot{S} = SZ \quad S(0) = S_0$$

where S_0 is some polynomial. The solution to the above differential equation is given by

$$S(t) = S_0 e^{tZ}$$

The exponential map of a polynomial and S_0 are both polynomials, hence their product $S(t)$ is also a polynomial.

From the definition of the exponential map, it is evident that,

$$e^X e^Y \neq e^Y e^X \neq e^{X+Y}$$

The product of exponential maps is provided by the Campbell-Baker-Hausdorff formula, which states,

$$e^X e^Y = e^{X+Y+\frac{1}{2}[X,Y]+\frac{1}{12}([X,[X,Y]]-[Y,[X,Y]])+\dots} \quad (\text{A.3})$$

The higher order terms can be obtained by expanding and multiplying the individual exponential maps.

With the above formula for product of exponentials, we have the following result. Let

$$G_k = \{e^Z | Z \in L_k(X_1, \dots, X_m)\}.$$

Then G_k is a *group* under multiplication. That is it satisfies the following properties: closed under multiplication, exist of inverse and identity element.

$$e^X e^Y = e^Z,$$

$$e^Z e^{-Z} = 1.$$

where $Z \in L_k(X_1, \dots, X_m)$. The above leads to a very important result. Any $S \in G_k$, can be expressed as a product of exponential maps of the P. Hall basis elements of $L_k(X_1, \dots, X_m)$, that is

$$S = e^{h_s B_s} \dots e^{h_1 B_1}, \quad (\text{A.4})$$

where $B_1, \dots, B_s$ are the P.Hall basis elements of $L_k(X_1, \dots, X_m)$ and h_i are P. Hall coordinates.

With these tools we proceed to discuss the motion planning algorithm followed by an example.

A.2 Algorithm

The implementation of this algorithm to a system in A.1 can be broadly divided into two steps:

1. Find a control v that steers the extended system for A.1 from the initial to the final configuration.
2. Use v to compute a control u that steers the actual system(A.1 from the same initial and to the same final configuration.

Step 1

Lets say the initial configuration of the system is $\mathcal{X}_e(0) = p$ and the final configuration is $\mathcal{X}_e(T) = q$.

- In order to find v , construct a $\mathcal{C}^1$ curve $\gamma : [0, T] \rightarrow \mathbb{R}^n$ that goes from p to q . A straight line segment is also possible.
- Once a suitable γ has been selected, its tangent vector $\dot{\gamma}(t)$ can be expressed as a linear combination of the vector fields of the extended system, that is $B_1(x), \dots, B_r(x)$. The resulting coefficients are the $v_i(t)$. These coefficients are found by computing the inverse (if $r = n$) or a pseudoinverse of a matrix whose column vectors are the control vector fields of the extended system.

$$\begin{bmatrix} \dot{X}_{1e}(t) \\ \vdots \\ \dot{X}_{ne}(t) \end{bmatrix} = \begin{bmatrix} B_1(\dot{X}_e(t)) & \dots & B_r(\dot{X}_e(t)) \end{bmatrix} \begin{bmatrix} v_1 \\ \vdots \\ v_r \end{bmatrix}$$

$$v(t) = [B]^{-1} \dot{X}_e$$

Step 2

The vector fields $g_1, \dots, g_m$ in A.1 belong to the vector space comprising of vector fields on the configuration manifold $\mathcal{M}$.

The noncommutative multiplication on these vector fields is defined as:

$$g_1 \cdot g_2 = [\nabla g_2](g_1) \quad (\text{A.5})$$

Therefore it can be observed that:

$$g_1 \cdot g_2(f) = \mathcal{L}_{g_1} \mathcal{L}_{g_2} f \neq g_2 \cdot g_1(f)$$

And,

$$[g_1, g_2] = [\nabla g_2](g_1) - [\nabla g_1](g_2)$$

The above satisfies the properties of Lie brackets.

The Lie algebra of the vector fields is given by $Lie^\infty(g_1, \dots, g_m)$. The nilpotent version is $Lie^k(g_1, \dots, g_m)$. This nilpotent Lie algebra will have a corresponding P. Hall basis $\mathcal{B}(m, k)$.

Vector field on a manifold gives rise to a differential equation. For a $Z \in Lie^\infty(g_1, \dots, g_m)$, we have,

$$\dot{x}(t) = Z(x) \quad x(0) = x_0$$

The solution to the above is given by Fliess expansion.

$$x(t) = x_0 + Z(x_0)t + (\nabla Z)(Z)(x_0) \frac{t^2}{2!} + \dots$$

From our definition of multiplication of vector fields in A.5, we obtain the following solution to the differential equation:

$$x(t) = [Id + Zt + Z^2 \frac{t^2}{2} + \dots](x_0) = e^{Zt} x_0$$

The exponential is essentially the flow along Z , that is, $\phi_t^Z = e^{Zt}$. Hence the concatenation $e^X e^Y$ is basically flow along X , followed by flow along Y and is given by the Campbell-Baker-Hausdorff formula in A.3. The same formula can be used to express flows along higher order brackets in terms of flows along individual vector fields, for example:

$$e^{\sqrt{t}X} e^{\sqrt{t}Y} e^{-\sqrt{t}X} e^{-\sqrt{t}Y} = e^{t[X,Y] + H.O.T} \quad (\text{A.6})$$

These higher order terms eventually vanish depending upon the order of nilpotency of the system.

Given some inputs $u_1, \dots, u_m$, they give rise to:

$$\dot{x}(t) = \sum_{j=1}^m g_j(x) u_j$$

and also give rise to this polynomial input:

$$Z = u_1 X_1 + \dots + u_m X_m$$

where X_i are indeterminates. Here Z is the input to the differential equation $\dot{S} = SZ$ evolving on a completely different space, that is the space of noncommutative polynomials. But the inputs in A.1, acts on a system evolving on a manifold. Essentially, these inputs are acting on two completely different spaces. However both these spaces have the same Lie algebraic structure. For the extended system in the polynomial space, the differential equation is

$$\dot{S}_e = S_e Z_e \quad (\text{A.7})$$

The Lie brackets in the extended systems for both spaces are the same. Provided this, it is now possible to estimate u corresponding to v of the extended system that drives the dynamical system from p to q . The v obtained from Step 1 is plugged into the extended polynomial system A.7. Using A.4, a general form of S , along with v is used to solve A.7. This will yield us ordinary differential equations of the P. Hall coordinates. The advantage of using backward P. Hall coordinates is that the functions $h_i(t)$ are easily computed by solving a system of O.D.Es with input v . This only requires successive integrations and algebraic operations. With the expressions for the P. Hall coordinates at hand, it is necessary to determine the S that achieves the same purpose as S_e . The fact that both of them have the same Lie algebras leads to them having identical expressions. However the higher order brackets that appear in S_e can be expressed in terms of individual original basis polynomials due to the Campbell-Baker-Hausdorff formula, as exhibited in A.6. This will provide us with the essential order of concatenation of flows and their respective time period. The above is precisely explained in the form of an example.

Example A.2.1 *This example illustrates the procedure to steer the system from an initial configuration $p = (0,0,0)$ to final configuration $q = (1,2,3)$ in 1 second. Consider the following dynamical system.*

$$\begin{bmatrix} \dot{q}_1(t) \\ \dot{q}_2(t) \\ \dot{q}_3(t) \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -q_2(t) \end{bmatrix} u_1 + \begin{bmatrix} 0 \\ 1 \\ q_1(t) \end{bmatrix} u_2 \quad (\text{A.8})$$

Construct a curve as follows:

$$\gamma(t) = \begin{bmatrix} t \\ 2t \\ 3t \end{bmatrix} \quad (\text{A.9})$$

The tangent vector of this curve is:

$$\dot{\gamma}(t) = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad (\text{A.10})$$

The vector fields are as follows:

$$g_1 = \begin{bmatrix} 1 \\ 0 \\ -q_2(t) \end{bmatrix} \quad g_2 = \begin{bmatrix} 0 \\ 1 \\ q_1(t) \end{bmatrix} \quad [g_1, g_2] = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} \quad [g_1, [g_1, g_2]] = [g_2, [g_1, g_2]] = 0$$

Therefore the above system is nilpotent of order 2. Since the number of vector fields is 2, we have $m = 2$ and $k = 2$. The P.Hall basis can be computed for this system as:

$$\mathcal{B}(2,2) = X, Y, [X, Y]$$

where $X \equiv g_1$ and $Y \equiv g_2$. Next step is to extend the system as below,

$$\begin{bmatrix} \dot{q}_1(t) \\ \dot{q}_2(t) \\ \dot{q}_3(t) \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -q_2(t) \end{bmatrix} v_1 + \begin{bmatrix} 0 \\ 1 \\ q_1(t) \end{bmatrix} v_2 + \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} v_3 \quad (\text{A.11})$$

From A.9 and A.10, the extended system can be rewritten as,

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2t & t & 2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \quad (\text{A.12})$$

Invert the above matrix to obtain the inputs to the extended system as:

$$v(t) = \begin{bmatrix} 1 \\ 2 \\ 1.5 \end{bmatrix}$$

Now applying this input to the extended polynomial system, we obtain:

$$\dot{S}_e = S_e \left(\sum_{j=1}^3 B_j v_j \right) \quad (\text{A.13})$$

Recalling that, any polynomial S can be expressed as a product of exponential maps of its basis polynomials, we obtain the solution to A.13 to be in the form of:

$$S_e = e^{h_3 B_3} e^{h_2 B_2} e^{h_1 B_1} = e^{h_3 [X, Y]} e^{h_2 Y} e^{h_1 X}$$

Substituting for S_e in A.13 and since the system is nilpotent of order 2 we get,

$$\begin{aligned} \frac{d}{dt} \left[(1 + h_3 [X, Y]) \left(1 + h_2 Y + \frac{h_2^2 Y^2}{2} \right) \left(1 + h_1 X + \frac{h_1^2 X^2}{2} \right) \right] &= [(1 + h_3 [X, Y]) \\ &\quad \left(1 + h_2 Y + \frac{h_2^2 Y^2}{2} \right) \left(1 + h_1 X + \frac{h_1^2 X^2}{2} \right)] (v_1 X + v_2 Y + v_3 [X, Y]) \end{aligned}$$

After equating the coefficients the following differential equations are obtained:

$$\begin{aligned} \dot{h}_1 &= v_1 \\ \dot{h}_2 &= v_2 \\ \dot{h}_3 &= v_3 + h_1 v_2 \end{aligned} \tag{A.14}$$

The above structure is independent of the problem. The value of v depends on the curve constructed. The final value of the coefficients are:

$$h(t) = \begin{bmatrix} t \\ 2t \\ t^2 + 1.5t \end{bmatrix}$$

However, we are only concerned with the values of these coefficients at time $T = 1s$, that is

$$h(T) = \begin{bmatrix} 1 \\ 2 \\ 2.5 \end{bmatrix}$$

Since the basis for the Lie algebra of the original and the extended system is the same, it results in $S = S_e$. Hence the final expression for the solution of the differential equation, S is:

$$S = e^{2.5[X,Y]} e^{2Y} e^X$$

From A.6, we can rewrite S as,

$$S = e^{\sqrt{2.5}X} e^{\sqrt{2.5}Y} e^{-\sqrt{2.5}X} e^{2 - \sqrt{2.5}X} e^X$$

u_1 corresponds to flow along X and u_2 corresponds to flow along Y . To reach q from p the following control inputs are applied: Since the total time taken for this maneuver is $1s$,

u_1	u_2	t
1	0	$t \in [0, \sqrt{2.5}]$
0	1	$t \in \sqrt{2.5}[1, 2]$
-1	0	$t \in \sqrt{2.5}[2, 3]$
0	1	$t \in [3\sqrt{2.5}, 2 + 2\sqrt{2.5}]$
1	0	$t \in [2 + 2\sqrt{2.5}, 3 + 2\sqrt{2.5}]$

we need to scale the final time with $\frac{1}{3+2\sqrt{2.5}}$ and scale the input values with $3 + 2\sqrt{2.5}$, to achieve our objective.

The example given above is of a nilpotent system. However this algorithm can be used to steer systems which aren't nilpotent as well. However, there is a lack of exact convergence to the final values but the error is bounded under certain conditions. The bound on this error is further discussed in detail in Lafferriere and Sussmann, 1993. For implementing it on systems which aren't nilpotent, better results are obtained for small step sizes in the configuration space. Now that the algorithm has been detailed, we can proceed to implement the same on the coordinated microscallop system in 4.33.

Appendix B

Lie bracket computation

B.1 Accessibility results in non Newtonian fluids

The Lie brackets computed for the accessibility analysis of a microscallop in non Newtonian fluids is listed below. These brackets correspond to the configuration of [3.30](#), that is $x = 0, y = 0, \theta = 0.92rad$ and $\alpha = 0.26rad$ with the scallop being at rest.

$$[f_0, f_1] = \begin{bmatrix} -2.43 \times 10^{10} \\ 0 \\ 0 \\ -1.21 \times 10^{14} \\ -5.25 \times 10^{10} \\ -3.32 \times 10^{11} \\ 1.66 \times 10^{15} \\ -9.84 \times 10^{13} \end{bmatrix} \quad [f_0, f_2] = \begin{bmatrix} 0 \\ 2.43 \times 10^{10} \\ 0 \\ 0 \\ 3.09 \times 10^{11} \\ 5.25 \times 10^{10} \\ -9.84 \times 10^{13} \\ 1.55 \times 10^{15} \end{bmatrix}$$

$$[f_0, [f_0, f_1]] = \begin{bmatrix} 5.25 \times 10^{10} \\ 3.32 \times 10^{11} \\ -1.66 \times 10^{15} \\ 9.84 \times 10^{13} \\ 3.92 \times 10^{12} \\ 7.06 \times 10^{11} \\ -1.29 \times 10^{15} \\ 2.15 \times 10^{16} \end{bmatrix} \quad [f_0, [f_0, f_2]] = \begin{bmatrix} -3.09 \times 10^{11} \\ -5.25 \times 10^{10} \\ 9.84 \times 10^{13} \\ -1.55 \times 10^{15} \\ -6.57 \times 10^{11} \\ -3.6 \times 10^{12} \\ 2 \times 10^{16} \\ 1.2 \times 10^{15} \end{bmatrix}$$

$$[f_1, [f_0, f_1]] = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ -2.15 \times 10^9 \\ 0 \\ 0 \end{bmatrix} \quad [f_1, [f_0, [f_0, f_1]]] = \begin{bmatrix} 0 \\ 1.85 \times 10^9 \\ 0 \\ 0 \\ -1.5 \times 10^{10} \\ -1.29 \times 10^{10} \\ -5.28 \times 10^{13} \\ -6.16 \times 10^{13} \end{bmatrix}$$

$$[f_1, [f_0, f_1]] = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 2.43 \times 10^{10} \\ 0 \\ 0 \\ 1.21 \times 10^{14} \end{bmatrix} \quad [f_1, [f_0, [f_0, f_1]]] = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ -2.43 \times 10^{10} \\ 1.21 \times 10^{14} \\ 0 \end{bmatrix}$$

The rank of the matrix obtained by appending these column vectors is 8. Hence the system is accessible from the concerned configuration.

B.2 STLC results in Newtonian fluids

The Lie brackets computed for the STLC analysis of the two microscallop system in Newtonian fluids is listed below. These brackets correspond to the configuration of 5.2, that is $x_A = 0$, $y_A = 0$, $\alpha_{1A} = 0.79rad$, $\alpha_{2A} = 1.32rad$, $x_B = 1$, $y_B = 2$, $\alpha_{1B} = 5.5rad$ and $\alpha_{2B} = 4.62rad$.

$$\begin{aligned}
 f_1 &= \begin{bmatrix} 0.000125515 \\ -0.000183867 \\ 1 \\ 0 \\ 7.96618 \times 10^{-13} \\ 2.48856 \times 10^{-12} \\ 0 \\ 0 \end{bmatrix} & f_2 &= \begin{bmatrix} 0.000222226 \\ -0.0000132995 \\ 0 \\ 1 \\ -1.56874 \times 10^{-12} \\ -2.24237 \times 10^{-12} \\ 0 \\ 0 \end{bmatrix} \\
 f_3 &= \begin{bmatrix} -6.9913 \times 10^{-13} \\ -3.91024 \times 10^{-12} \\ 0 \\ 0 \\ -0.000137974 \\ -0.000203377 \\ 1 \\ 0 \end{bmatrix} & f_4 &= \begin{bmatrix} 1.20304 \times 10^{-12} \\ -1.0814 \times 10^{-13} \\ 0 \\ 0 \\ -0.000236226 \\ 0.0000677961 \\ 0 \\ 1 \end{bmatrix} \\
 [f_1, f_2] &= \begin{bmatrix} -0.000083608 \\ 0.0000474052 \\ 0 \\ 0 \\ 2.57317 \times 10^{-12} \\ -1.68451 \times 10^{-12} \\ 0 \\ 0 \end{bmatrix} & [f_2, f_3] &= \begin{bmatrix} 7.47244 \times 10^{-16} \\ 5.17184 \times 10^{-16} \\ 0 \\ 0 \\ 3.11718 \times 10^{-16} \\ 5.16234 \times 10^{-16} \\ 0 \\ 0 \end{bmatrix} \\
 [f_3, f_4] &= \begin{bmatrix} 6.10522 \times 10^{-12} \\ 4.62809 \times 10^{-13} \\ 0 \\ 0 \\ -0.000126799 \\ -0.0000459424 \\ 0 \\ 0 \end{bmatrix} & [f_1, f_3] &= \begin{bmatrix} 8.29853 \times 10^{-16} \\ 1.16125 \times 10^{-15} \\ 0 \\ 0 \\ -1.24333 \times 10^{-16} \\ -5.27655 \times 10^{-16} \\ 0 \\ 0 \end{bmatrix}
 \end{aligned}$$

The rank of the matrix obtained by appending these column vectors is 8. Hence the system is STLC from the concerned configuration.

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